CONVERGENCE OF THE GLOWINSKI DECOMPOSITION–COORDINATION METHOD FOR THE g-LAPLACIAN: ASSUMPTIONS ON THE SOLUTION, THE DISCRETE PROBLEM, THE LAVRENTIEV PHENOMENON, AND NUMERICAL EXPERIMENTS

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Abstract. We study the decomposition–coordination (augmented Lagrangian, ALG2/ADMM) method of Glowinski for the Dirichlet problem $-\operatorname{div}\big(g(x,|\nabla u|)\nabla u/|\nabla u|\big) = f$, i.e. for the minimisation of $\int_\Omega G(x,|\nabla v|)\,dx - \int_\Omega f v\,dx$, where $G(x,\cdot)$ is an arbitrary convex Young function. At the continuous level, in $H^1_0 \times L^2 \times L^2$ and with the multiplier step equal to the penalty parameter, we prove: (A) weak convergence of the iterates and convergence of the residual to zero, using only convexity of G, under the assumption (S) that the augmented Lagrangian has a saddle point; (B) strong convergence in H^1_0 when, in addition, $G(x,\cdot) \in C^1$, $g(x,0) = 0$ and $g(x,t)/t$ is nondecreasing in t and bounded below by a positive function of t uniformly in x, and in $W^{1,p}_0$ for the p-Laplacian with $1 < p < 2$. We show that the hypotheses of the classical convergence theorem fail for $p < 2$, and we review, with their exact hypotheses, the regularity results that imply (S), the cases where they are not available (exponential growth, double phase up to the boundary) and the known counterexamples. For the finite element problem (P1 elements, piecewise constant fluxes) we prove (C) that the iteration converges for every penalty parameter and every convex G with superlinear growth, without any regularity assumption. We discuss the consequences of the Lavrentiev phenomenon for conforming discretisations, and for the autonomous radial test problems we prove convergence in energy of the P1 minimisers. Finally we report experiments with exact solutions, with errors computed by quadrature: the observed orders are 2 in L^2 and 1 for the gradient for smooth G, which goes beyond what is proved; we also check the Lyapunov identity of the proof numerically and study the influence of the penalty parameter.

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1. Introduction

Let $\Omega \subset \mathbb{R}^d$ be a bounded open set and let $G\colon \Omega \times [0,\infty) \to [0,\infty)$ be such that $G(x,\cdot)$ is a convex Young function for a.e. x. We are interested in the numerical solution of

$$\min_v \; J(v), \qquad J(v) = \int_\Omega G(x,|\nabla v|)\,dx - \int_\Omega f v\,dx, \tag{1.1}$$

with homogeneous Dirichlet data, whose Euler–Lagrange equation is the *g-Laplace equation*

$$-\operatorname{div}\Big(g(x,|\nabla u|)\frac{\nabla u}{|\nabla u|}\Big) = f \quad \text{in } \Omega, \qquad u = 0 \quad \text{on } \partial\Omega, \qquad g = \partial_t G. \tag{1.2}$$

This class contains the p-Laplacian ($G = t^p/p$), the $p(x)$-Laplacian ($G = t^{p(x)}/p(x)$) that appears in models of electrorheological fluids [2], the double phase and two-phase energies $t^p/p + a(x)t^q/q$ with $a \geq 0$ [52, 22], Orlicz energies such as $t^p \log^a(1+t)$, and energies of exponential growth such as $e^t - 1 - t$, which do not satisfy the Δ_2 condition.

The decomposition–coordination method of Glowinski and Marrocco [40], Fortin and Glowinski [35, 36] and Glowinski [39] (the algorithm "ALG2"; for the choice of parameters made here it belongs to the family of alternating direction methods of multipliers, ADMM [41, 14]) introduces the flux variable $\nu = \nabla u$ and alternates a *linear* Poisson problem for u, a *pointwise* nonlinear problem for ν, and an update of the multiplier η. For the $p(x)$-Laplacian it was implemented with P1 finite elements by Aragón, Fernández Bonder and Rubio [4]; the only place where G enters the algorithm is a scalar monotone equation in each element, so the same code works for an arbitrary Young function. The open-source package `glaplace` that we use for all experiments does exactly this (see the paragraph *Software availability* at the end of the paper).

The convergence theory of the method is classical in Hilbert spaces [39, 35, 41]. Its hypotheses, as reproduced in [41], include that the functional $F(q) = \int_\Omega G(x,|q|)\,dx$ be superlinear on L^2 and contain a Gâteaux differentiable part that is uniformly convex on bounded sets of L^2. For $G = t^p/p$ with $1 < p < 2$ these hypotheses fail (Proposition 2.13); the convergence theorem in [4] is stated for $p(x) \geq 2$. (For $p > 2$ the functional is not finite on L^2, so the differentiability hypothesis has to be understood on $\operatorname{dom} F$; we comment on this in Section 2.6.) The method, on the other hand, is

routinely used for $p < 2$ and for functions G of non-polynomial growth. The purpose of this paper is to clarify what can be proved, under which assumptions, at the continuous and at the discrete level, and to complement this with careful numerical experiments.

Contributions.

(1) *Weak convergence at the continuous level (Theorem A).* For every convex Young integrand $G(x,t)$, without growth or differentiability conditions, and for the multiplier step equal to the penalty parameter ($\rho = r$), if the augmented Lagrangian has a saddle point in $H^1_0 \times L^2 \times L^2$ (Assumption S), then the iterates converge weakly to a saddle point, $u_n \to u$ strongly in L^2 and the residual $\nabla u_n - \nu_n$ tends to 0 in L^2. The proof is self-contained: an explicit Lyapunov identity (2.9) and an Opial-type argument.

(2) *Strong convergence (Theorem B)* under (S), in H^1_0 if $G(x,\cdot) \in C^1$, $g(x,0) = 0$, and $g(x,t)/t$ is nondecreasing in t and bounded below by a positive function of t uniformly in x ("superquadratic type", including $e^t - 1 - t$ and two-phase energies with $2 \le p \le q$), and in $W^{1,p}_0$ for the p-Laplacian with $1 < p < 2$, a range not covered by the classical theorem.

(3) *Discussion of the assumption (S)* (Section 3), which is an assumption on the solution: the regularity results that imply it, with their exact hypotheses, the cases in which no such result is available, and the known counterexamples (Table 1).

(4) *Convergence of the discrete iteration (Theorem C)* for every $r > 0$ and every convex G with superlinear growth (including t^p/p for all $p > 1$ and $e^t - 1$), with no regularity assumption: in finite dimension a saddle point always exists, and the proof of Theorem A applies verbatim.

(5) *The Lavrentiev phenomenon* (Section 5): when it occurs, the conforming discrete problem need not approximate the continuous minimiser. For our radial test problems we show that the exact minimiser is Lipschitz, so no conforming function has lower energy (Lemma 5.1), and for the autonomous ones we prove convergence in energy of the P1 minimisers (Proposition 5.3).

(6) *Numerical experiments* (Section 6) with exact solutions and true (quadrature) errors, a numerical verification of the Lyapunov identity, and a study of the penalty parameter.

We stress what is and is not new. The proof of Theorem A is a careful but standard Fejér/Opial argument (compare [49, 11]), and Theorem C(iii) is standard finite-dimensional ADMM theory [31]; we include complete proofs because the setting (arbitrary Young integrands, nonsmooth F, no growth conditions) and the exact hypotheses matter for the discussion. The new points are Theorem B(b), the characterisation of (S) for the p-Laplacian (Proposition 2.15), the existence of discrete saddle points without any regularity (Theorem C(i)–(ii)), and the systematic discussion of (S) and of the Lavrentiev phenomenon in relation to the method.

What is not proved. We emphasise five limitations. (i) Assumption S is a regularity property of the solution; it is available from the literature only in some cases (Section 3), and convergence of the continuous iteration when it fails (for instance for $1 < p < 2$ when the minimiser is not in H^1_0) is open. (ii) We do not prove any rate, neither for the iteration nor for the finite element error; the orders reported in Section 6 are observations. (iii) We treat only $\rho = r$. (iv) We do not prove, in general, that the P1 minimisers u_h converge to the continuous minimiser as $h \to 0$; Lemma 5.1 gives only the lower bound $J(v_h) \ge J(u)$ for conforming v_h, and Proposition 5.3 covers only the autonomous radial test problems (not the two-phase ones). (v) Proposition 2.15 and Section 4 are written for $d = 2$ (Proposition 2.15 holds more generally for $p \ge 2d/(d+2)$).

Notation. $(\cdot,\cdot)$ and $\|\cdot\|$ denote the inner product and norm of $L^2(\Omega)$ or $L^2(\Omega;\mathbb{R}^d)$; other norms carry a subscript. For a convex function $\Psi\colon X \to (-\infty, +\infty]$ on a Hilbert space, $\operatorname{dom}\Psi = \{\Psi < \infty\}$ and $\partial\Psi(q) = \{\zeta \in X : \Psi(q') \ge \Psi(q) + (\zeta, q' - q)\ \forall q' \in X\}$ for $q \in \operatorname{dom}\Psi$. For a convex function γ on $[0,\infty)$, γ'_+ denote its one-sided derivatives. Weak convergence is denoted by $\rightharpoonup$.

2. The method at the continuous level

2.1. Setting.

Definition 2.1 (Young integrand)**.** A function $G\colon \Omega \times [0,\infty) \to [0,\infty)$ is a *Young integrand* if $G(\cdot,t)$ is measurable for every $t \geq 0$ and, for a.e. $x \in \Omega$, $G(x,\cdot)$ is finite, convex and nondecreasing with $G(x,0) = 0$.

A finite convex nondecreasing function on $[0,\infty)$ vanishing at 0 is continuous (at 0: $0 \leq G(x,t) \leq (t/s)G(x,s)$ for $0 < t < s$), so G is a Carathéodory function and $x \mapsto G(x,|q(x)|)$ is measurable for every measurable q. We write

$$\Phi(x,\xi) = G(x,|\xi|), \qquad g(x,t) = \partial_t^+ G(x,t) \quad (t \geq 0),$$

so that $\Phi(x,\cdot)$ is convex on $\mathbb{R}^d$ (composition of a convex nondecreasing function with a norm) and $g(x,\cdot)$ is nondecreasing, finite and nonnegative; $g(x,0) = \inf_{s>0} G(x,s)/s$. No growth condition ($\Delta_2$, ∇_2, superlinearity) is assumed in this section.

Let $V = H_0^1(\Omega)$ with the norm $\|v\|_V = \|\nabla v\|$, $H = L^2(\Omega;\mathbb{R}^d)$, $f \in L^2(\Omega)$, and

$$F(q) = \int_\Omega \Phi(x,q(x))\,dx \in [0,+\infty], \qquad q \in H; \qquad J(v) = F(\nabla v) - (f,v), \quad v \in V.$$

Lemma 2.2. *F is convex, $F \geq 0 = F(0)$, and F is lower semicontinuous on H; hence also weakly lower semicontinuous.*

Proof. Convexity and $F \geq 0 = F(0)$ are clear. If $q_k \to q$ in H, choose a subsequence with $F(q_{k_j}) \to \liminf_k F(q_k)$ and $q_{k_j} \to q$ a.e.; by continuity of $\Phi(x,\cdot)$ and Fatou's lemma $F(q) \leq \liminf_j F(q_{k_j})$. The sublevel sets of F are convex and closed, hence weakly closed (Mazur), which is weak lower semicontinuity. □

Abstract structure. To make clear what the proofs use, we write $B = \nabla\colon V \to H$ and $\ell(v) = (f,v)$. The results of Sections 2.2–2.4 use only the following properties:

(A1) V and H are real Hilbert spaces and $B\colon V \to H$ is linear with $\|Bv\|_H = \|v\|_V$;
(A2) $F\colon H \to [0,+\infty]$ is convex, lower semicontinuous and proper ($\operatorname{dom} F \neq \emptyset$);
(A3) $\ell \in V^*$.

The only additional ingredients in the concrete case are the compactness of $H_0^1(\Omega) \hookrightarrow L^2(\Omega)$ (Rellich's theorem, valid for every bounded open set) and the pointwise description of ∂F (Lemma 2.4). The same abstract structure is used in Section 4 for the finite element problem.

2.2. Augmented Lagrangian and saddle points.

For $r > 0$ define on $V \times H \times H$

$$\mathcal{L}_r(v,q,\mu) = F(q) - \ell(v) + (\mu, Bv - q) + \frac{r}{2}\|Bv - q\|^2.$$

A triple (u^*,ν^*,η^*) is a saddle point if $\mathcal{L}_r(u^*,\nu^*,\mu) \leq \mathcal{L}_r(u^*,\nu^*,\eta^*) \leq \mathcal{L}_r(v,q,\eta^*)$ for all (v,q,μ), with the middle term finite.

Proposition 2.3 (Saddle points)**.** *Assume (A1)–(A3). $(u^*,\nu^*,\eta^*) \in V \times H \times H$ is a saddle point of $\mathcal{L}_r$ if and only if*

$$\nu^* = Bu^*, \qquad \eta^* \in \partial F(\nu^*), \qquad (\eta^*, Bw) = \ell(w) \quad \forall w \in V. \tag{2.1}$$

In that case $F(\nu^*) < \infty$ and u^* minimises $J(v) = F(Bv) - \ell(v)$ over V. In particular the set of saddle points does not depend on r.

Proof. ($\Leftarrow$) Since $Bu^* = \nu^*$, $\mathcal{L}_r(u^*, \nu^*, \mu) = F(\nu^*) - \ell(u^*)$ does not depend on μ, and it is finite because $\eta^* \in \partial F(\nu^*)$ requires $\nu^* \in \operatorname{dom} F$. For $(v, q) \in V \times H$, the subgradient inequality $F(q) \geq F(\nu^*) + (\eta^*, q - \nu^*)$ and (2.1) with $w = v - u^*$ give

$$\begin{aligned}\mathcal{L}_r(v, q, \eta^*) - \mathcal{L}_r(u^*, \nu^*, \eta^*) &\geq (\eta^*, q - \nu^*) - \ell(v - u^*) + (\eta^*, Bv - q) + \frac{r}{2}\|Bv - q\|^2 \\ &= (\eta^*, B(v - u^*)) - \ell(v - u^*) + \frac{r}{2}\|Bv - q\|^2 \geq 0,\end{aligned}$$

where we used $\nu^* = Bu^*$.

($\Rightarrow$) The middle value is finite, so $F(\nu^*) < \infty$. The left inequality, $(\mu, Bu^* - \nu^*) \leq (\eta^*, Bu^* - \nu^*)$ for all $\mu \in H$, forces $Bu^* = \nu^*$. The right inequality with $(v, q) = (u^* + sw, \nu^*)$, $w \in V$, $s \in \mathbb{R}$, reads $s\big[(\eta^*, Bw) - \ell(w)\big] + \frac{rs^2}{2}\|Bw\|^2 \geq 0$; dividing by $s > 0$ and by $s < 0$ and letting $s \to 0$ gives $(\eta^*, Bw) = \ell(w)$. The right inequality with $v = u^*$ and $q = \nu^* + s(q' - \nu^*)$, $q' \in \operatorname{dom} F$, $s \in (0, 1)$, together with convexity, gives

$$F(\nu^*) \leq F(\nu^*) + s\big(F(q') - F(\nu^*)\big) - s(\eta^*, q' - \nu^*) + \frac{rs^2}{2}\|q' - \nu^*\|^2;$$

dividing by s and letting $s \downarrow 0$ yields $F(q') \geq F(\nu^*) + (\eta^*, q' - \nu^*)$, trivially true also if $q' \notin \operatorname{dom} F$. Hence $\eta^* \in \partial F(\nu^*)$.

Finally, if (2.1) holds, then for $v \in V$: $J(v) - J(u^*) \geq (\eta^*, B(v - u^*)) - \ell(v - u^*) = 0$. Condition (2.1) does not involve r. □

Assumption S. The augmented Lagrangian $\mathcal{L}_r$ has a saddle point in $V \times H \times H$; equivalently, (2.1) has a solution.

In the concrete case, (S) says that J has a minimiser $u \in H^1_0(\Omega)$ whose *flux* η (an element of $\partial F(\nabla u)$, see Lemma 2.4) belongs to $L^2(\Omega; \mathbb{R}^d)$ and satisfies the weak Euler–Lagrange equation $(\eta, \nabla w) = (f, w)$ for all $w \in H^1_0$. It is a property of the solution, not only of G; Section 3 discusses when it is known to hold.

Lemma 2.4 (Pointwise description of ∂F)**.** *Let $q \in \operatorname{dom} F$ and $\eta \in H$. Then $\eta \in \partial F(q)$ if and only if $\eta(x) \in \partial\Phi(x, \cdot)(q(x))$ for a.e. $x \in \Omega$.*

Proof. ($\Leftarrow$) If $q' \in \operatorname{dom} F$, the pointwise inequality $\Phi(x, q'(x)) \geq \Phi(x, q(x)) + \eta(x) \cdot (q'(x) - q(x))$ involves integrable functions and can be integrated; if $q' \notin \operatorname{dom} F$ there is nothing to prove.

($\Rightarrow$) Fix $z \in \mathbb{Q}^d$ and $M \in \mathbb{N}$, let $\Omega^z_M = \{x : \Phi(x, z) \leq M\}$ and let $A \subset \Omega^z_M$ be measurable. The field $q' = q\,\mathbf{1}_{\Omega\setminus A} + z\,\mathbf{1}_A$ belongs to $H \cap \operatorname{dom} F$, and $F(q') \geq F(q) + (\eta, q' - q)$ reads

$$\int_A \big[\Phi(x, z) - \Phi(x, q(x)) - \eta(x) \cdot (z - q(x))\big]\, dx \geq 0.$$

The integrand is integrable on Ω^z_M and A is arbitrary, so it is ≥ 0 a.e. on Ω^z_M. Since $\Phi(x, z) < \infty$, $\bigcup_M \Omega^z_M = \Omega$ up to a null set. Taking the union of the exceptional null sets over $z \in \mathbb{Q}^d$ we find a null set N such that $\Phi(x, z) \geq \Phi(x, q(x)) + \eta(x) \cdot (z - q(x))$ for $x \notin N$ and all $z \in \mathbb{Q}^d$, hence, by continuity of $\Phi(x, \cdot)$, for all $z \in \mathbb{R}^d$. □

Lemma 2.5 (Subdifferential of Φ)**.** *Let $\gamma\colon [0, \infty) \to [0, \infty)$ be finite, convex, nondecreasing, $\gamma(0) = 0$, and $\phi(\xi) = \gamma(|\xi|)$ on $\mathbb{R}^d$.*

(i) $\partial\phi(0) = \{\zeta : |\zeta| \leq \gamma'_+(0)\}$.
(ii) *If $\xi \neq 0$ and γ is differentiable at $|\xi|$, then $\partial\phi(\xi) = \{\gamma'(|\xi|)\xi/|\xi|\}$.*
(iii) *If $\gamma \in C^1([0, \infty))$ with $\gamma'(0) = 0$, then $\phi \in C^1(\mathbb{R}^d)$, $\nabla\phi(\xi) = T(\xi) := \gamma'(|\xi|)\xi/|\xi|$ ($T(0) = 0$), and $\partial\phi(\xi) = \{T(\xi)\}$ for every ξ.*

Proof. (i) $\zeta \in \partial\phi(0)$ iff $\gamma(|z|) \geq \zeta \cdot z$ for all z, iff $\gamma(t) \geq |\zeta| t$ for all $t \geq 0$, iff $|\zeta| \leq \inf_{t>0} \gamma(t)/t = \gamma'_+(0)$ (the difference quotient of a convex function is nondecreasing). (ii) and (iii): a convex function

differentiable at ξ has $\partial\phi(\xi) = \{\nabla\phi(\xi)\}$: the gradient is a subgradient by convexity, and if $\zeta \in \partial\phi(\xi)$ then $(\phi(\xi + sz) - \phi(\xi))/s \geq \zeta \cdot z$ for $s > 0$, so $\nabla\phi(\xi) \cdot z \geq \zeta \cdot z$ for all z, i.e. $\zeta = \nabla\phi(\xi)$. At $\xi \neq 0$ the chain rule gives $\nabla\phi(\xi) = \gamma'(|\xi|)\xi/|\xi|$. In (iii), $\phi(\xi) = \gamma(|\xi|) = o(|\xi|)$ at 0 because $\gamma(0) = \gamma'(0) = 0$, so $\nabla\phi(0) = 0$, and T is continuous at 0 because $\gamma'(t) \to 0$. □

In particular, when $G(x, \cdot) \in C^1([0, \infty))$ with $g(x, 0) = 0$, Lemmas 2.4 and 2.5 give, for $q \in \operatorname{dom} F$,

$$\partial F(q) = \{T(\cdot, q)\} \text{ if } T(\cdot, q) \in H, \qquad \partial F(q) = \emptyset \text{ otherwise;} \qquad T(x, \xi) = g(x, |\xi|)\frac{\xi}{|\xi|}. \tag{2.2}$$

For $G(x, t) = e^t - 1$, instead, $g(x, 0) = 1$ and $\partial\Phi(x, 0)$ is the closed unit ball: ∂F is multivalued at fields that vanish on a set of positive measure.

2.3. **The algorithm.** Fix $r > 0$, $\nu_0 \in H$ and $\eta_1 \in \partial F(\nu_0)$. If moreover $F(0) = 0$ (as for the F of Section 2.1), then $0 \in \partial F(0)$ because $F \geq 0$, and $\nu_0 = \eta_1 = 0$ is admissible; this is the initialisation used in the code. By Remark 2.9 the condition $\eta_1 \in \partial F(\nu_0)$ is not needed anyway. For $n \geq 1$:

$$u_n \in V: \quad (\eta_n + r(Bu_n - \nu_{n-1}), Bw) = \ell(w) \quad \forall w \in V, \tag{2.3}$$

$$\nu_n = \operatorname*{argmin}_{q \in H} \Big\{ F(q) - (\eta_n, q) + \frac{r}{2}\|Bu_n - q\|^2 \Big\}, \tag{2.4}$$

$$\eta_{n+1} = \eta_n + r(Bu_n - \nu_n). \tag{2.5}$$

Step (2.3) minimises $\mathcal{L}_r(\cdot, \nu_{n-1}, \eta_n)$, step (2.4) minimises $\mathcal{L}_r(u_n, \cdot, \eta_n)$, and (2.5) is a multiplier step of length $\rho = r$. This is ALG2 of [36, 39] with $\rho = r$, in the form used in [4]. In the concrete case (2.3) is the Poisson problem $-r\Delta u_n = f + \operatorname{div}(\eta_n - r\nu_{n-1})$, $u_n \in H_0^1$, with the same operator at every iteration. We write

$$s_n = Bu_n - \nu_n \qquad \text{(primal residual)}, \qquad \text{so that} \quad \eta_{n+1} - \eta_n = r s_n.$$

Lemma 2.6 (Well-posedness)**.** *Assume (A1)–(A3).*

(i) (2.3) *has a unique solution* u_n.

(ii) (2.4) *has a unique solution, and* $\nu \in H$ *is that solution if and only if* $\eta_n + r(Bu_n - \nu) \in \partial F(\nu)$. *Consequently*

$$\eta_{n+1} \in \partial F(\nu_n) \qquad (n \geq 0). \tag{2.6}$$

(iii) *For* $n \geq 1$,

$$(\eta_{n+1} + r(\nu_n - \nu_{n-1}), Bw) = \ell(w) \qquad \forall w \in V. \tag{2.7}$$

Proof. (i) By (A1), $(v, w) \mapsto r(Bv, Bw)$ is r times the inner product of V, and $w \mapsto \ell(w) - (\eta_n - r\nu_{n-1}, Bw)$ is in V^*; apply the Riesz theorem.

(ii) Let $\Psi_n(q) = F(q) - (\eta_n, q) + \frac{r}{2}\|Bu_n - q\|^2$. It is proper (since $\operatorname{dom} F \neq \emptyset$), convex, lower semicontinuous, hence weakly lower semicontinuous, and coercive: $\Psi_n(q) \geq -\|\eta_n\|\|q\| + \frac{r}{2}(\|q\| - \|Bu_n\|)^2$ because $F \geq 0$. Bounded sets are weakly sequentially compact, so a minimiser exists; it is unique because Ψ_n is strictly convex. If ν minimises Ψ_n, then for $q \in \operatorname{dom} F$ and $s \in (0, 1)$, with $\nu_s = \nu + s(q - \nu)$,

$$0 \leq \Psi_n(\nu_s) - \Psi_n(\nu) \leq s\big(F(q) - F(\nu)\big) - s(\eta_n, q - \nu) - rs(Bu_n - \nu, q - \nu) + \frac{rs^2}{2}\|q - \nu\|^2;$$

dividing by s and letting $s \downarrow 0$ gives $F(q) \geq F(\nu) + (\eta_n + r(Bu_n - \nu), q - \nu)$, i.e. $\eta_n + r(Bu_n - \nu) \in \partial F(\nu)$. Conversely, if this inclusion holds, adding the subgradient inequality to the identity $\frac{r}{2}\|Bu_n - q\|^2 - \frac{r}{2}\|Bu_n - \nu\|^2 = -r(Bu_n - \nu, q - \nu) + \frac{r}{2}\|q - \nu\|^2$ shows $\Psi_n(q) \geq \Psi_n(\nu)$. With $\nu = \nu_n$, (2.5) gives (2.6) for $n \geq 1$; for $n = 0$ it is the choice of η_1.

(iii) By (2.5), $\eta_n + r(Bu_n - \nu_{n-1}) = \eta_{n+1} + r(\nu_n - \nu_{n-1})$; insert in (2.3). □

The second step decouples into independent problems at each point. The following lemma describes them; it is exactly what the code implements (a bisection for a scalar monotone equation).

Lemma 2.7 (The pointwise problem)**.** *Let γ be as in Lemma 2.5, $\phi(\xi)=\gamma(|\xi|)$, $r>0$ and $w\in\mathbb{R}^d$.*

(i) *$\xi\mapsto\phi(\xi)+\frac{r}{2}|\xi-w|^2$ has a unique minimiser $\hat\xi=\hat t\,w/|w|$ ($\hat\xi=0$ if $w=0$), where $\hat t\in[0,|w|]$ is the unique minimiser of $\chi(t)=\gamma(t)+\frac{r}{2}(t-|w|)^2$ on $[0,\infty)$.*

(ii) *$\hat t=0$ if and only if $r|w|\le\gamma'_+(0)$; otherwise $\hat t>0$ is characterised by $\gamma'_-(\hat t)\le r(|w|-\hat t)\le\gamma'_+(\hat t)$.*

(iii) *If $\gamma\in C^1$ and $\gamma'(0)<r|w|$, then $\hat t$ is the unique root in $(0,|w|]$ of the strictly increasing function $t\mapsto\gamma'(t)+rt-r|w|$.*

(iv) *$\hat\eta:=r(w-\hat\xi)\in\partial\phi(\hat\xi)$, and $|\hat\eta|=r(|w|-\hat t)$.*

Proof. (i) The function χ is continuous, strictly convex and coercive on $[0,\infty)$, so it has a unique minimiser $\hat t$; for $t>|w|$, $\chi(t)>\chi(|w|)$ because γ is nondecreasing, so $\hat t\le|w|$. For every ξ, $|\xi-w|\ge\big||\xi|-|w|\big|$, hence $\phi(\xi)+\frac{r}{2}|\xi-w|^2\ge\chi(|\xi|)\ge\chi(\hat t)$, and the value $\chi(\hat t)$ is attained at $\hat\xi$. Uniqueness follows from strict convexity. (ii) For a convex function on $[0,\infty)$, $\hat t$ is a minimiser iff $\chi'_+(\hat t)\ge0$ and, if $\hat t>0$, $\chi'_-(\hat t)\le0$; and $\chi'_\pm(t)=\gamma'_\pm(t)+r(t-|w|)$. (iii) follows from (ii) and the strict monotonicity of $\gamma'(t)+rt$. (iv) If $w=0$, $\hat\eta=0\in\partial\phi(0)$. If $\hat t=0<|w|$, $|\hat\eta|=r|w|\le\gamma'_+(0)$ and Lemma 2.5(i) applies. If $\hat t>0$, then for every z, using $|z|\ge\hat w\cdot z$ with $\hat w=w/|w|$, the monotonicity of γ and the convexity inequality $\gamma(s)\ge\gamma(\hat t)+\lambda(s-\hat t)$ with $\lambda=r(|w|-\hat t)\in[\gamma'_-(\hat t),\gamma'_+(\hat t)]$ (valid for $s\ge0$),

$$\phi(z)\ge\gamma(\max\{\hat w\cdot z,0\})\ge\gamma(\hat t)+\lambda\big(\hat w\cdot z-\hat t\big)=\phi(\hat\xi)+\hat\eta\cdot(z-\hat\xi),$$

where in the case $\hat w\cdot z<0$ we used $\gamma(0)=0\ge\gamma(\hat t)+\lambda(\hat w\cdot z-\hat t)$, which follows from the convexity inequality at $s=0$ and $\lambda\ge0$. □

Corollary 2.8 (Pointwise form of (2.4))**.** *In the concrete case, let $w_n=\eta_n/r+\nabla u_n$. For a.e. x, $\nu_n(x)$ is the minimiser of Lemma 2.7 with $\gamma=G(x,\cdot)$ and $w=w_n(x)$: $\nu_n(x)=t(x)\,w_n(x)/|w_n(x)|$, where $t(x)=0$ if $r|w_n(x)|\le g(x,0)$ and otherwise, when $G(x,\cdot)\in C^1$, $t(x)\in(0,|w_n(x)|]$ solves*

$$g(x,t)+r\,t=r|w_n(x)|. \tag{2.8}$$

Moreover $\eta_{n+1}(x)=r(w_n(x)-\nu_n(x))\in\partial\Phi(x,\cdot)(\nu_n(x))$; when $\nu_n(x)\neq0$ and $G(x,\cdot)$ is differentiable at $|\nu_n(x)|$, $\eta_{n+1}(x)=g(x,|\nu_n(x)|)\nu_n(x)/|\nu_n(x)|$: the multiplier is the flux.

Proof. By (2.6) and Lemma 2.4, $\zeta(x):=\eta_{n+1}(x)=r(w_n(x)-\nu_n(x))\in\partial\Phi(x,\cdot)(\nu_n(x))$ a.e. Adding the subgradient inequality to $\frac{r}{2}|z-w|^2-\frac{r}{2}|\xi-w|^2\ge r(\xi-w)\cdot(z-\xi)$ shows that $\xi=\nu_n(x)$ minimises $\Phi(x,\cdot)+\frac{r}{2}|\cdot-w_n(x)|^2$; by uniqueness it is the minimiser of Lemma 2.7. The rest is Lemma 2.7 and Lemma 2.5(ii). □

2.4. **Theorem A: weak convergence under (S).**

Theorem A. *Assume (A1)–(A3) and Assumption S. Let (u_n,ν_n,η_n) be generated by (2.3)–(2.5), let (u^*,ν^*,η^*) be any saddle point, and set*

$$E_n=\frac{1}{2r}\|\eta_{n+1}-\eta^*\|^2+\frac{r}{2}\|\nu_n-\nu^*\|^2\qquad(n\ge0).$$

Then for every $n\ge1$

$$E_n+\frac{r}{2}\|s_n\|^2+\frac{r}{2}\|\nu_n-\nu_{n-1}\|^2+(\eta_{n+1}-\eta^*,\nu_n-\nu^*)+r(s_n,\nu_n-\nu_{n-1})=E_{n-1}, \tag{2.9}$$

and the last two terms on the left are nonnegative. Consequently (E_n) is nonincreasing; (η_n), (ν_n) are bounded in H and (u_n) in V;

$$\sum_{n\ge1}\|s_n\|^2<\infty,\qquad\sum_{n\ge1}\|\nu_n-\nu_{n-1}\|^2<\infty,\qquad\sum_{n\ge1}m_n<\infty,\quad m_n:=(\eta_{n+1}-\eta^*,\nu_n-\nu^*)\ge0;$$

in particular $Bu_n - \nu_n \to 0$, $\nu_n - \nu_{n-1} \to 0$ *and* $\eta_{n+1} - \eta_n \to 0$ *strongly in* H. *Moreover there is a saddle point* $(\bar{u}, \bar{\nu}, \bar{\eta})$ *such that*

$$u_n \rightharpoonup \bar{u} \text{ in } V, \qquad \nu_n \rightharpoonup \bar{\nu}, \quad \eta_n \rightharpoonup \bar{\eta} \text{ in } H.$$

In the concrete case $V = H_0^1(\Omega)$: $u_n \rightharpoonup \bar{u}$ *weakly in* $H_0^1(\Omega)$ *and strongly in* $L^2(\Omega)$, $\nabla u_n - \nu_n \to 0$ *in* $L^2(\Omega;\mathbb{R}^d)$, *and* $\bar{u}$ *minimises* J *over* $H_0^1(\Omega)$; *if this minimiser is unique (for instance if* $G(x,\cdot)$ *is strictly convex for a.e.* x*),* $\bar{u}$ *is the unique minimiser.*

No growth condition and no differentiability of G is used. Theorem A applies, whenever (S) holds, to t^p/p for every $p > 1$, to $p(x)$, p-q and two-phase energies, to $t^p \log^a(1+t)$, to $e^t - 1 - t$ and to $e^t - 1$ (where ∂F is multivalued).

Proof. Step 1: the identity (2.9). Fix $n \geq 1$. Subtract $(\eta^*, Bw) = \ell(w)$ from (2.7) and take $w = u_n - u^*$; since $\nu^* = Bu^*$, $Bw = s_n + (\nu_n - \nu^*)$, and we obtain

$$(\eta_{n+1} - \eta^*, s_n) + (\eta_{n+1} - \eta^*, \nu_n - \nu^*) + r(\nu_n - \nu_{n-1}, s_n) + r(\nu_n - \nu_{n-1}, \nu_n - \nu^*) = 0.$$

By the identity $2(a-c, a-b) = \|a-c\|^2 + \|a-b\|^2 - \|b-c\|^2$ and $rs_n = \eta_{n+1} - \eta_n$,

$$(\eta_{n+1} - \eta^*, s_n) = \frac{1}{r}(\eta_{n+1} - \eta^*, \eta_{n+1} - \eta_n) = \frac{1}{2r}\big(\|\eta_{n+1} - \eta^*\|^2 - \|\eta_n - \eta^*\|^2\big) + \frac{r}{2}\|s_n\|^2,$$
$$r(\nu_n - \nu_{n-1}, \nu_n - \nu^*) = \frac{r}{2}\big(\|\nu_n - \nu^*\|^2 - \|\nu_{n-1} - \nu^*\|^2\big) + \frac{r}{2}\|\nu_n - \nu_{n-1}\|^2.$$

Inserting these two identities gives (2.9). Note that Step 1 uses only (2.7), $\nu^* = Bu^*$ and $(\eta^*, Bw) = \ell(w)$.

Step 2: signs and consequences. By (2.6), (2.1) and the monotonicity of ∂F (add the two subgradient inequalities), $m_n = (\eta_{n+1} - \eta^*, \nu_n - \nu^*) \geq 0$. Also $\eta_n \in \partial F(\nu_{n-1})$ for $n \geq 1$, so $r(s_n, \nu_n - \nu_{n-1}) = (\eta_{n+1} - \eta_n, \nu_n - \nu_{n-1}) \geq 0$. Thus $E_n \leq E_{n-1}$, and summing (2.9) from 1 to N gives $E_N + \sum_{n=1}^N \big(\frac{r}{2}\|s_n\|^2 + \frac{r}{2}\|\nu_n - \nu_{n-1}\|^2 + m_n\big) \leq E_0$. Hence (η_{n+1}) and (ν_n) are bounded, the three series converge, and $Bu_n = s_n + \nu_n$ is bounded, i.e. (u_n) is bounded in V by (A1).

Step 3: weak cluster points are saddle points. Let $x_n = (\nu_n, \eta_{n+1}) \in H \times H$ and let $(\bar{\nu}, \bar{\eta})$ be a weak cluster point: $\nu_{n_k} \rightharpoonup \bar{\nu}$, $\eta_{n_k+1} \rightharpoonup \bar{\eta}$; passing to a further subsequence, $u_{n_k} \rightharpoonup \bar{u}$ in V. Since B is linear and bounded, $Bu_{n_k} \rightharpoonup B\bar{u}$; since $Bu_n - \nu_n \to 0$, $B\bar{u} = \bar{\nu}$. In (2.7) with $n = n_k$ the term $r(\nu_n - \nu_{n-1}, Bw)$ tends to 0, so $(\bar{\eta}, Bw) = \ell(w)$ for all $w \in V$. Taking $w = u_n$ in (2.7) and using $Bu_n = s_n + \nu_n$,

$$(\eta_{n+1}, \nu_n) = \ell(u_n) - (\eta_{n+1}, s_n) - r(\nu_n - \nu_{n-1}, Bu_n) \xrightarrow[n=n_k\to\infty]{} \ell(\bar{u}) = (\bar{\eta}, B\bar{u}) = (\bar{\eta}, \bar{\nu}),$$

because the last two terms are products of a bounded and a strongly null sequence and $\ell(u_{n_k}) \to \ell(\bar{u})$. Now, by (2.6), for every $q \in H$,

$$F(q) \geq F(\nu_{n_k}) + (\eta_{n_k+1}, q) - (\eta_{n_k+1}, \nu_{n_k}).$$

Letting $k \to \infty$ and using the weak lower semicontinuity of F (Lemma 2.2 in the concrete case; in general (A2) and Mazur's lemma) we get $F(q) \geq F(\bar{\nu}) + (\bar{\eta}, q - \bar{\nu})$. Taking $q = \nu^*$ shows $F(\bar{\nu}) < \infty$, so $\bar{\eta} \in \partial F(\bar{\nu})$. By Proposition 2.3, $(\bar{u}, \bar{\nu}, \bar{\eta})$ is a saddle point.

Step 4: convergence of the whole sequence. Let $\mathcal{S}$ be the set of pairs (ν^*, η^*) such that (u^*, ν^*, η^*) is a saddle point for some $u^* \in V$, and equip $H \times H$ with the inner product $\langle (\nu, \eta), (\nu', \eta') \rangle = r(\nu, \nu') + \frac{1}{r}(\eta, \eta')$ and norm $|\cdot|$. Then $|x_n - z|^2 = 2E_n(z)$ for $z \in \mathcal{S}$, which converges by Step 2, and every weak cluster point of the bounded sequence (x_n) lies in $\mathcal{S}$ by Step 3. Let z_1, z_2 be two weak cluster points. Since

$$|x_n - z_1|^2 - |x_n - z_2|^2 = -2\langle x_n, z_1 - z_2 \rangle + |z_1|^2 - |z_2|^2$$

converges, $\langle x_n, z_1 - z_2 \rangle$ converges; evaluating the limit along the two subsequences gives $\langle z_1, z_1 - z_2 \rangle = \langle z_2, z_1 - z_2 \rangle$, i.e. $|z_1 - z_2|^2 = 0$. Hence $x_n \rightharpoonup \bar{z} = (\bar{\nu}, \bar{\eta}) \in \mathcal{S}$, and $\eta_n \rightharpoonup \bar{\eta}$ since $\eta_{n+1} - \eta_n \to 0$.

Finally $Bu_n = s_n + \nu_n \rightharpoonup \bar\nu$; every weak cluster point $\bar u$ of the bounded sequence (u_n) satisfies $B\bar u = \bar\nu$, and B is injective, so $u_n \rightharpoonup \bar u$ in V, and $(\bar u, \bar\nu, \bar\eta)$ is a saddle point by Step 3.

In the concrete case, Rellich's theorem upgrades the weak convergence in $H^1_0(\Omega)$ to strong convergence in $L^2(\Omega)$, and Proposition 2.3 shows that $\bar u$ minimises J. □

Remark 2.9 (On the initialisation). The condition $\eta_1 \in \partial F(\nu_0)$ is used only for the sign of the cross term at $n = 1$. Without it, (2.9) still holds and gives $E_1 \le E_0 + |r(s_1, \nu_1 - \nu_0)|$; all conclusions remain true, since for $n \ge 2$ the argument is unchanged.

Remark 2.10 (Related results). The quantity E_n is, up to normalisation, the Lyapunov functional used for ADMM with equal penalty and multiplier step by Bartels and Milicevic [11, Thm. 3.7] (their D_j^2, which equals $2rE$ for $\tau = r$), and, as far as we recall, in the finite-dimensional analysis of Boyd et al. [14, App. A]. Weak convergence of ADMM in infinite-dimensional Hilbert spaces is also obtained in [49] through the equivalence between ADMM and the Douglas–Rachford method applied to the dual problem [37, 41] and the weak convergence theorems for Douglas–Rachford [45, 31, 51, 12]. Our proof works directly with the primal–dual iterates and the identity (2.9), which is used again in Theorem B and is checked numerically in Section 6.6.

Remark 2.11 (Weak, not strong, convergence). For a general convex F we only obtain weak convergence of (ν_n, η_n) and of u_n in V. This is the general situation: in infinite dimensions the Douglas–Rachford algorithm converges only weakly, and Bùi and Combettes [16] give an example (with two maximally monotone operators in a separable Hilbert space) in which its iterates converge weakly but not strongly. Since ADMM with $\rho = r$ is Douglas–Rachford applied to the dual problem, the general ADMM/Douglas–Rachford theory gives only weak convergence; strong convergence requires additional structure such as uniform monotonicity (Theorem B). We do not know whether strong convergence can fail for some instance of problem (1.1).

2.5. **Theorem B: strong convergence.** Assume now that $G(x,\cdot) \in C^1([0,\infty))$ with $g(x,0) = 0$ for a.e. x, and let $T(x,\xi) = g(x,|\xi|)\xi/|\xi|$ and

$$k(x,t) = \frac{g(x,t)}{t}\ (t>0), \qquad \psi(x;a,b) = (T(x,a) - T(x,b))\cdot(a-b) \ge 0.$$

By (2.2) and (2.6), $\eta_{n+1} = T(\cdot,\nu_n)$ and $\eta^* = T(\cdot,\nu^*)$ a.e. for every saddle point, hence

$$m_n = \int_\Omega \psi(x;\nu_n(x),\nu^*(x))\,dx. \tag{2.10}$$

Lemma 2.12. (i) *If $t \mapsto k(x,t)$ is nondecreasing on $(0,\infty)$ and we set $k(x,0) = \lim_{t\downarrow 0} k(x,t)$, then for all $a, b \in \mathbb{R}^d$*

$$\psi(x;a,b) = \tfrac12\big(k(x,|a|)+k(x,|b|)\big)|a-b|^2 + \tfrac12\big(k(x,|a|)-k(x,|b|)\big)\big(|a|^2-|b|^2\big) \ \ge\ \tfrac12\, k\big(x,\tfrac{|a-b|}{2}\big)|a-b|^2.$$

(ii) *Let $T(\xi) = |\xi|^{p-2}\xi$ with $1 < p \le 2$. Then for all $a, b \in \mathbb{R}^d$*

$$(T(a)-T(b))\cdot(a-b) \ge (p-1)(|a|+|b|)^{p-2}|a-b|^2 \quad (a,b \text{ not both } 0), \qquad |T(a)-T(b)| \le 2^p|a-b|^{p-1}.$$

Proof. (i) Write $k_a = k(x,|a|)$, $k_b = k(x,|b|)$, so $T(x,a) = k_a a$ (also for $a = 0$). Expanding, both ψ and the right-hand side of the identity equal $k_a|a|^2 + k_b|b|^2 - (k_a + k_b)\,a\cdot b$. The second term is ≥ 0 because $k(x,\cdot)$ is nondecreasing on $[0,\infty)$. Hence $\psi \ge \frac12(k_a+k_b)|a-b|^2 \ge \frac12 k(x,\max\{|a|,|b|\})|a-b|^2$, and $\max\{|a|,|b|\} \ge |a-b|/2$.

(ii) T is C^1 on $\mathbb{R}^d\setminus\{0\}$ with $DT(\xi) = |\xi|^{p-2}\big(I + (p-2)\hat\xi\otimes\hat\xi\big)$, $\hat\xi = \xi/|\xi|$, a symmetric matrix with eigenvalues $|\xi|^{p-2}$ and $(p-1)|\xi|^{p-2}$. Let $c_s = b + s(a-b)$, $s \in [0,1]$. The segment meets 0 at most at one s_0 (unless $a = b$, a trivial case), $s \mapsto T(c_s)$ is continuous on $[0,1]$, C^1 off s_0, and $|\frac{d}{ds}T(c_s)| \le |c_s|^{p-2}|a-b|$, which is integrable because $|c_s| \ge |a-b||s-s_0|$ if the segment meets 0

and $p-2>-1$. Hence $s\mapsto T(c_s)$ is absolutely continuous and

$$(T(a)-T(b))\cdot(a-b)=\int_0^1 (a-b)^\top DT(c_s)(a-b)\,ds$$
$$\geq (p-1)|a-b|^2\int_0^1 |c_s|^{p-2}ds\geq (p-1)(|a|+|b|)^{p-2}|a-b|^2,$$

since $|c_s|\leq |a|+|b|$ and $p-2\leq 0$. For the Hölder bound let $\delta=|a-b|$ and assume $|a|\geq|b|$. If $|a|\leq 2\delta$, then $|T(a)-T(b)|\leq |a|^{p-1}+|b|^{p-1}\leq 2(2\delta)^{p-1}\leq 2^p\delta^{p-1}$. If $|a|>2\delta$, then $|c_s|\geq |a|-\delta>\delta$ on the whole segment, the operator norm of $DT(c_s)$ is $|c_s|^{p-2}\leq\delta^{p-2}$, and $|T(a)-T(b)|\leq\delta\cdot\delta^{p-2}=\delta^{p-1}$. □

Theorem B. *Assume (S) and that $G(x,\cdot)\in C^1([0,\infty))$ with $g(x,0)=0$ for a.e. x. Let $(\bar u,\bar\nu,\bar\eta)$ be the limit in Theorem A.*

(a) (Superquadratic type.) *Suppose that for a.e. x the function $t\mapsto k(x,t)=g(x,t)/t$ is nondecreasing on $(0,\infty)$, and that $k(x,t)\geq k_0(t)>0$ for all $t>0$ and a.e. x, for some function k_0. Then $\nu_n\to\bar\nu$ strongly in $L^2(\Omega;\mathbb{R}^d)$ and $u_n\to\bar u$ strongly in $H^1_0(\Omega)$. Moreover $\nabla\bar u=\nu^*$ for every saddle point, so the minimiser of J over $H^1_0(\Omega)$ is unique.*

(b) (Sub-quadratic power.) *Let $G(x,t)=t^p/p$ with $1<p<2$. Then $\nu_n\to\bar\nu$ and $\nabla u_n\to\nabla\bar u$ strongly in $L^s(\Omega;\mathbb{R}^d)$ for every $s<2$, in particular $u_n\to\bar u$ strongly in $W^{1,p}_0(\Omega)$; moreover $\eta_n\to\bar\eta$ strongly in $L^2(\Omega;\mathbb{R}^d)$, and $u_n\to\bar u$ in $L^2(\Omega)$.*

Proof. Apply Theorem A with the saddle point (u^*,ν^*,η^*); by (2.10), $\int_\Omega\psi(x;\nu_n,\nu^*)\,dx=m_n\to 0$. Put $e_n=\nu_n-\nu^*$.

(a) Let $\delta>0$. By Lemma 2.12(i), the monotonicity of $k(x,\cdot)$ and the hypothesis at the single value $t=\delta/2$, for a.e. x in the set $\{|e_n|>\delta\}$ we have $\psi(x;\nu_n,\nu^*)\geq\frac12 k(x,|e_n|/2)|e_n|^2\geq\frac12 k(x,\delta/2)|e_n|^2\geq\frac12 k_0(\delta/2)|e_n|^2$, and $\psi\geq 0$ everywhere, so

$$\|e_n\|^2\leq\delta^2|\Omega|+\frac{2}{k_0(\delta/2)}\,m_n.$$

Hence $\limsup_n\|e_n\|^2\leq\delta^2|\Omega|$ for every $\delta>0$, i.e. $\nu_n\to\nu^*$ in L^2. Since also $\nu_n\rightharpoonup\bar\nu$, $\nu^*=\bar\nu$ for every saddle point, and $\nabla u_n=s_n+\nu_n\to\bar\nu=\nabla\bar u$ in L^2. Finally, every minimiser u' of J on H^1_0 belongs to a saddle point: if $J(u')=J(u^*)$, the subgradient inequality and (2.1) give $F(\nabla u')=F(\nabla u^*)+(\eta^*,\nabla(u'-u^*))$, hence $F(q)\geq F(\nabla u^*)+(\eta^*,q-\nabla u^*)=F(\nabla u')+(\eta^*,q-\nabla u')$ for all q, i.e. $(u',\nabla u',\eta^*)$ satisfies (2.1). Therefore $\nabla u'=\bar\nu$ and $u'=\bar u$.

(b) By Lemma 2.12(ii) and Hölder's inequality with exponents $2/p$ and $2/(2-p)$,

$$\int_\Omega|e_n|^p\,dx\leq\Big(\int_\Omega(|\nu_n|+|\nu^*|)^{p-2}|e_n|^2\,dx\Big)^{p/2}\Big(\int_\Omega(|\nu_n|+|\nu^*|)^p\,dx\Big)^{1-p/2}\leq\Big(\frac{m_n}{p-1}\Big)^{p/2}C,$$

where the integrand of the first factor is set to 0 where $\nu_n=\nu^*=0$, and C bounds the second factor because (ν_n) is bounded in $L^2\subset L^p$. Thus $\nu_n\to\nu^*$ in L^p; since $\nu_n\rightharpoonup\bar\nu$, $\nu^*=\bar\nu$. As (ν_n) is bounded in L^2, the interpolation inequality $\|e\|_{L^s}\leq\|e\|_{L^p}^{\theta}\|e\|_{L^2}^{1-\theta}$ $(p<s<2)$ gives convergence in L^s for all $s<2$; the same holds for $\nabla u_n=s_n+\nu_n$, and Poincaré's inequality in $W^{1,p}_0$ gives $u_n\to\bar u$ in $W^{1,p}_0$. Finally, $\eta_{n+1}=T(\nu_n)$, $\bar\eta=T(\bar\nu)$, and by Lemma 2.12(ii) and Hölder's inequality (note $2(p-1)\leq p$)

$$\|\eta_{n+1}-\bar\eta\|^2\leq 4^p\int_\Omega|e_n|^{2(p-1)}\,dx\leq 4^p|\Omega|^{1-\frac{2(p-1)}{p}}\|e_n\|_{L^p}^{2(p-1)}\to 0.$$ □

Case (a) contains t^p/p for $p\geq 2$, $t^p/p+a(x)t^q/q$ with $2\leq p\leq q$ and measurable $a\geq 0$ (here $k\geq t^{p-2}$; a may be discontinuous), $t^2\log(1+t)$ (for which $k(t)=2\log(1+t)+t/(1+t)$), $t^3/\log(1+t)$, and e^t-1-t (for which $k(t)=(e^t-1)/t\geq 1$), all without Δ_2/∇_2 conditions. Since $\partial_t(g/t)=(g'-g/t)/t$, for $G\in C^2$ the monotonicity of k is equivalent to $tg'(t)/g(t)\geq 1$. For t^p/p,

$p \geq 2$, (a) is the strong convergence in H_0^1 that [4] derive from Glowinski's theorem. Case (b) is not covered by that theorem (Section 2.6); note that it gives strong convergence of ν_n in L^s, $s < 2$, not in L^2. We do not claim strong convergence of η_n in case (a).

2.6. **Relation with Glowinski's theorem; the restriction $p \geq 2$.** We have not been able to consult [39]; the following summary of the hypotheses of the convergence theorems for ALG1 and ALG2 is taken from Glowinski, Pan and Tai [41, §3.2], the most complete accessible reproduction (its numbering in [39], Ch. VI, Thms. 4.1 and 5.1, is the one used in [4, 26]). For $\min_{v\in V} F(Bv)+G_2(v)$ with V, H Hilbert spaces, $B \in \mathcal{L}(V,H)$ and F, G_2 proper, convex and lower semicontinuous, assume

(i) the augmented Lagrangian has a saddle point;
(ii) B is injective and $B(V)$ is closed in H;
(iii) $F(q)/\|q\| \to +\infty$ as $\|q\| \to \infty$;
(iv) $F = F_0 + F_1$ with F_0, F_1 proper, convex, lower semicontinuous, and F_0 Gâteaux differentiable and uniformly convex on the bounded sets of H.

Then for ALG2 with $0 < \rho < \frac{1+\sqrt{5}}{2} r$, $(u_n, \nu_n) \to (u, Bu)$ in $V \times H$ (strongly, as we read [41]), while the multipliers converge, in general, only weakly [41]. (In [41] the two minimisation steps of ALG2 are written in the order "ν-step, u-step".) In [4, Thm. 4.3] this theorem is applied with $V = H_0^1$, $H = L^2$, $G_2 = -\ell$ and $F(q) = \int |q|^{p(x)}/p(x)$, for $p(x) \geq 2$. For $1 < p < 2$ the hypotheses fail:

Proposition 2.13. *Let $F(q) = \frac{1}{p}\int_\Omega |q|^p\,dx$ on $H = L^2(\Omega;\mathbb{R}^d)$ with $1 < p < 2$. Then F satisfies neither* (iii) *nor* (iv)*: $F(q)/\|q\| \not\to \infty$, and F is not uniformly convex on the unit ball of H (and F_0 cannot be split off as in* (iv)*, see the proof).*

Proof. Fix a unit vector e, and for M large a measurable $A_M \subset \Omega$ with $|A_M| = M^{-2+\varepsilon}$, $0 < \varepsilon < 2(2-p)$; let $q_M = M\mathbf{1}_{A_M} e$. Then $\|q_M\| = M^{\varepsilon/2} \to \infty$ and $F(q_M)/\|q_M\| = M^{p-2+\varepsilon/2}/p \to 0$, so (iii) fails. Next take $|A_M| = M^{-2}$, so $\|q_M\| = 1$; then

$$\frac{F(0)+F(q_M)}{2} - F\Big(\frac{q_M}{2}\Big) = \frac{M^{p-2}}{p}\Big(\frac{1}{2} - 2^{-p}\Big) \to 0, \qquad \|q_M - 0\| = 1.$$

If $F = F_0 + F_1$ with F_1 convex and F_0 uniformly convex on the unit ball with modulus $\delta(\cdot)$, the left-hand side would be $\geq \delta(1) > 0$ for all M (the corresponding expression for F_1 is ≥ 0); contradiction. □

For $p > 2$ the situation is not clear-cut either: $F = +\infty$ on $L^2 \setminus L^p$, so F is not Gâteaux differentiable on H, and $F - \frac{c}{2}\|\cdot\|^2$ is not convex near 0 for any $c > 0$ (along $q = se$, e a constant unit field, it equals $|\Omega|(s^p/p - cs^2/2)$), so there is no obvious splitting $F = F_0 + F_1$ as in (iv) with F_0 differentiable on all of H. Hypothesis (iv) must therefore be understood in a weaker sense (for instance with F_0 differentiable on $\operatorname{dom} F$) for the theorem to apply to $p > 2$, as it is applied in [4]; we have not been able to check this in [39]. What we can say with certainty is that the hypotheses as reproduced above fail for $p < 2$.

Thus for $1 < p < 2$ the classical theorem does not apply, although, by Theorems A and B(b), the iteration converges under (S). Two further differences: Theorem A requires neither (iii) nor the differentiability in (iv), so it covers $e^t - 1$; and Theorem A gives only weak convergence for general F, in agreement with Remark 2.11: it is the uniform convexity in (iv) (or the uniform monotonicity used in Theorem B) that yields strong convergence, while injectivity and closed range of B only transfer the type of convergence of Bu_n to u_n. We have treated only $\rho = r$; for $\rho \neq r$ the Lyapunov functional must be modified, and this is where the bound $\rho < \frac{1+\sqrt{5}}{2} r$ of Fortin and Glowinski appears [36, 41].

2.7. **What (S) means for the p-Laplacian.** For $G = t^p/p$ the meaning of (S) depends on whether $p < 2$ or $p > 2$. We use the following abstract multiplier lemma, which is also the key to Theorem C.

Lemma 2.14 (Multiplier lemma)**.** *Let X be a real Hilbert space, $R \subset X$ a closed subspace, and $\Psi\colon X \to \mathbb{R}$ convex and bounded above on a ball centred at $q_0 \in R$. If q_0 minimises Ψ over R, then there is $\zeta \in \partial\Psi(q_0)$ with $\zeta \perp R$.*

Proof. For $z \in X$ the difference quotient $s \mapsto (\Psi(q_0+sz)-\Psi(q_0))/s$ is nondecreasing on $s>0$ and bounded below by $\Psi(q_0)-\Psi(q_0-z)$, so $\Psi'(q_0;z) = \lim_{s\downarrow 0}(\Psi(q_0+sz)-\Psi(q_0))/s$ is finite; $z \mapsto \Psi'(q_0;z)$ is positively homogeneous and, by convexity, subadditive; and $\Psi'(q_0;z) \le \Psi(q_0+z)-\Psi(q_0)$. Since q_0 minimises Ψ on R, $\Psi'(q_0;z) \ge 0$ for $z \in R$. By the Hahn–Banach theorem (extension dominated by a sublinear functional) the zero functional on R extends to a linear functional ζ on X with $\zeta(z) \le \Psi'(q_0;z) \le \Psi(q_0+z)-\Psi(q_0)$ for all z. If $\Psi \le M$ on the ball of radius δ about q_0, then $|\zeta(z)| \le M - \Psi(q_0)$ for $\|z\| \le \delta$, so ζ is continuous; identifying it with an element of X, $\zeta \perp R$ and $\Psi(q_0+z) \ge \Psi(q_0) + (\zeta, z)$, i.e. $\zeta \in \partial\Psi(q_0)$. □

Proposition 2.15. *Let $G(t) = t^p/p$ and $d = 2$ (more generally $d \ge 2$ and $p \ge 2d/(d+2)$, so that $W_0^{1,p}(\Omega) \hookrightarrow L^2(\Omega)$). Let $u_p \in W_0^{1,p}(\Omega)$ be the unique minimiser of J over $W_0^{1,p}(\Omega)$.*

(i) *If $1 < p < 2$:* (S) *holds* $\iff$ *J attains its minimum on $H_0^1(\Omega)$* $\iff$ *$u_p \in H_0^1(\Omega)$.*

(ii) *If $p > 2$ and $\nabla u_p \in L^{2p-2}(\Omega)$, then* (S) *holds. Conversely, if Ω is a Lipschitz domain and* (S) *holds, then $\nabla u_p \in L^{2p-2}(\Omega)$.*

Proof. (i) If (S) holds, J attains its minimum on $V = H_0^1$ at u^* (Proposition 2.3). If J attains its minimum on V at u, apply Lemma 2.14 with $X = H$, $R = \nabla(V)$ (closed, ∇ being an isometry), $\lambda \in R$ the Riesz representative of $\nabla w \mapsto (f, w)$ on R, $\Psi(q) = F(q) - (\lambda, q)$ and $q_0 = \nabla u$; Ψ is finite and bounded on balls because $|q|^p \le 1 + |q|^2$. We get $\zeta \in \partial\Psi(\nabla u)$, $\zeta \perp R$; then $\eta = \zeta + \lambda \in \partial F(\nabla u)$ and $(\eta, \nabla w) = (\lambda, \nabla w) = (f, w)$ for $w \in V$, so $(u, \nabla u, \eta)$ satisfies (2.1). Next, V is dense in $W_0^{1,p}$ (it contains C_c^∞) and J is continuous on $W_0^{1,p}$ (here we use $W_0^{1,p} \hookrightarrow L^2$). If u minimises J on V, then $J(v) = \lim J(v_k) \ge J(u)$ for $v \in W_0^{1,p}$ and $v_k \in V$, $v_k \to v$ in $W^{1,p}$, so $u = u_p$ by uniqueness, i.e. $u_p \in V$. Conversely, if $u_p \in V$ it minimises J over $V \subset W_0^{1,p}$.

(ii) The Euler–Lagrange equation $(\eta, \nabla w) = (f, w)$, $\eta = |\nabla u_p|^{p-2}\nabla u_p$, holds for $w \in W_0^{1,p}$, in particular for $w \in C_c^\infty$. If $\nabla u_p \in L^{2p-2}$ then $\eta \in L^2$, both sides are continuous on V, and by density the equation holds on V; moreover $u_p \in W_0^{1,p} \subset V$, $F(\nabla u_p) < \infty$ and $\eta \in \partial F(\nabla u_p)$ by Lemmas 2.4 and 2.5. For the converse, let (u^*, ν^*, η^*) be a saddle point. Then $F(\nabla u^*) < \infty$, so $\nabla u^* \in L^p$; on a Lipschitz domain a function of H_0^1 with gradient in L^p has zero trace and belongs to $W^{1,p}(\Omega)$, hence to $W_0^{1,p}(\Omega)$. Since u^* minimises J on $V \supset W_0^{1,p}$, $u^* = u_p$, and $\eta^* = |\nabla u_p|^{p-2}\nabla u_p \in L^2$ by (2.2), i.e. $\nabla u_p \in L^{2p-2}$. □

So for $p < 2$ the function F is finite and continuous on L^2 and (S) is equivalent to $u_p \in H_0^1$; for $p > 2$, $F = +\infty$ on $L^2 \setminus L^p$, $u_p \in H_0^1$ is automatic, and (S) is a higher integrability requirement on ∇u_p (not implied by $u_p \in W_0^{1,p}$). In both ranges (S) is an assumption on the solution. When it fails, for instance for $p < 2$ and $u_p \notin H_0^1$, the iterates $u_n \in H_0^1$ are still well defined, but our argument says nothing; convergence in that case is open.

3. When does (S) hold? Regularity, its limits, and counterexamples

3.1. Two sufficient conditions.

Lemma 3.1 (Lipschitz solutions)**.** *Assume that $G(x,\cdot) \in C^1([0,\infty))$ with $g(x,0) = 0$ for a.e. x, and that $\operatorname{ess\,sup}_x g(x,M) < \infty$ for every $M > 0$. If $u \in H_0^1(\Omega)$ has $\nabla u \in L^\infty(\Omega)$ and satisfies $\int_\Omega T(x,\nabla u)\cdot\nabla\varphi\,dx = \int_\Omega f\varphi\,dx$ for all $\varphi \in C_c^\infty(\Omega)$, then (S) holds with $(u, \nabla u, T(\cdot,\nabla u))$.*

Proof. $\eta = T(\cdot,\nabla u)$ is measurable and bounded, hence in H; both sides of the equation are continuous on V, so by density it holds for all $w \in V$. $F(\nabla u) < \infty$, and $\eta \in \partial F(\nabla u)$ by Lemmas 2.4 and 2.5(iii). □

Lemma 3.2 (Explicit flux)**.** *Let $u \in H_0^1(\Omega)$ with $F(\nabla u) < \infty$ and $\eta \in H$ with $\eta(x) \in \partial\Phi(x,\cdot)(\nabla u(x))$ a.e. and $(\eta, \nabla w) = (f, w)$ for all $w \in H_0^1(\Omega)$ (for instance $\eta \in W^{1,\infty}(\Omega;\mathbb{R}^d)$ with $-\operatorname{div}\eta = f$ a.e.). Then (S) holds.*

Proof. Lemma 2.4 and Proposition 2.3; for $\eta \in W^{1,\infty}$, $(\eta, \nabla w) = -(\operatorname{div}\eta, w)$ for $w \in C_c^\infty$, hence for $w \in H_0^1$ by density. □

Lemma 3.2 applies to all radial test problems of Section 6, *including* the energies not covered by any regularity theory below; see Remark 5.2. Note also that (S) does not require a bounded gradient (Proposition 2.15).

3.2. **Regularity results that give (S).** We now list results from the literature that, combined with Lemma 3.1 (or with Proposition 2.15), yield (S). We state them as faithfully as we could verify them; items whose hypotheses we could not check in the original are explicitly marked, and the conclusion (S) is conditional on them. Throughout, d denotes the dimension.

Literature statement 3.3 (Autonomous Orlicz-type operators; Lieberman, Antonini)**.** Let $G = G(t)$, $g = G' \in C^1(0,\infty)$, $g(0) = 0$, with

$$0 < \delta \le \frac{t\,g'(t)}{g(t)} \le g_0 < \infty \qquad (t > 0). \tag{3.1}$$

Lieberman [44] developed the regularity theory for this structure condition (which for t^p/p reads $\delta = g_0 = p - 1$ and implies Δ_2 and ∇_2); in his theory g depends on t only, and x may enter only through a Hölder perturbation of the vector field of size $\Lambda(1+g(|\xi|))|x-y|^\theta$. His Theorem 1.7 is cited in the literature for $C^{1,\beta}(\overline{\Omega})$ regularity of bounded weak solutions with bounded right-hand side, $\partial\Omega \in C^{1,\alpha}$ and $C^{1,\alpha}$ Dirichlet data; we have not verified the boundary part in the original, and it has also been attributed to a different paper of the same author (see [3]). A verified global statement is Antonini's [3, Cor. 1.4]: for $-\operatorname{div} A(x,\nabla u) = f$ whose ξ-derivative is controlled by $a(|\xi|) = b(|\xi|)/|\xi|$ with $-1 < i_a \le s_a < \infty$ (equivalent to (3.1) for $b = g$), with $|A(x,\xi) - A(y,\xi)| \le \Lambda(1+b(|\xi|))|x-y|^\alpha$ for a *single* function b, on a bounded $C^{1,\alpha}$ domain with $f \in L^\sigma$, $\sigma > d$, and $C^{1,\alpha}$ Dirichlet data, one has $u \in C^{1,\beta}(\overline{\Omega})$.

Consequently (S) holds, by Lemma 3.1, for autonomous G satisfying (3.1) on bounded $C^{1,\alpha}$ domains with $f \in L^\sigma$, $\sigma > d$ (for example the disk with constant f). Among our test functions, (3.1) holds for t^p/p ($p > 1$), $t^2/2 + t^4/4$ ($tg'/g \in [1,3]$), $t^{3/2}/(3/2) + t^3/3$ ($tg'/g \in [\frac12, 2]$), and for $t^2\log(1+t)$ ($tg'/g \in (1,2]$) and $t^3/\log(1+t)$ ($tg'/g \in [1,2)$), as one checks from the asymptotics at 0 and ∞ and the monotonicity of tg'/g (verified numerically on $[10^{-8}, 10^8]$). For these two functions also $tg'/g \ge 1$, i.e. $g(t)/t$ is nondecreasing.

Literature statement 3.4 (Convex domains; Cianchi–Maz'ya)**.** Consider autonomous operators $-\operatorname{div}(a(|\nabla u|)\nabla u) = f$ with $-1 < i_a \le s_a < \infty$, where $i_a = \inf ta'(t)/a(t)$ and $s_a = \sup ta'(t)/a(t)$.

(a) on convex domains (no further boundary regularity) or domains with boundary in $W^2L^{d-1,1}$, and $f \in L^{d,1}$, the gradient is globally bounded [19] (possibly only for $d \ge 3$); the version for p-type a with $2 \le p < d$, $d \ge 3$, is [20, Thm. 3.5], and [20] shows that boundedness of the gradient may fail without these domain assumptions. (For [19] we have only the abstract; the precise hypotheses, including the range of d, must be checked.)

(b) On convex domains with $f \in L^2$, $a(|\nabla u|)\nabla u \in W^{1,2}(\Omega)$, with two-sided bounds in terms of $\|f\|$ [21, Thm. 2.3].

Statement (b) gives (S) on convex plane domains for every autonomous G of this class, in particular for t^p/p with every $p > 1$, provided the solution considered in [21] is the weak solution in $W_0^{1,G}$ (to be checked): indeed $\eta = a(|\nabla u|)\nabla u \in W^{1,2}(\Omega) \subset L^s(\Omega)$ for all $s < \infty$ when $d = 2$; since $b(t) = ta(t) \ge a(1)t^{1+i_a}$ for $t \ge 1$, also $\nabla u \in L^s$ for all $s < \infty$, so $u \in H_0^1$, and the Euler–Lagrange

equation extends to H_0^1 by density; Lemma 2.4 gives $\eta \in \partial F(\nabla u)$. This is the natural route to (S) for $1 < p < 2$ (Proposition 2.15(i)).

Literature statement 3.5 (Variable exponent)**.** For $G = t^{p(x)}/p(x)$ with $p \in C^{0,\mu}(\overline{\Omega})$, $\inf p > 1$, $\partial\Omega \in C^{1,\beta}$, bounded weak solutions of the Dirichlet problem are in $C^{1,\alpha}(\overline{\Omega})$, according to Fan [33] as cited by many authors; we have not read the hypotheses in the original. Interior $C^{1,\beta}$ regularity for Hölder exponents is due to Coscia–Mingione [24] and Acerbi–Mingione [1]; log-Hölder continuity of p gives Hölder continuity of u and density of smooth functions (no Lavrentiev gap; this is attributed to Zhikov, cf. [53]), but not, by itself, a bounded gradient.

Under Fan's hypotheses (S) holds by Lemma 3.1 ($g(x, M) \le \max\{1, M\}^{\sup p-1}$).

Literature statement 3.6 (Double phase: interior results only)**.** Consider the double phase energy $\int(|\nabla w|^p + a(x)|\nabla w|^q)$, $1 < p \le q$, $0 \le a \in C^{0,\alpha}$. Local minimisers have locally Hölder continuous gradient if $q/p < 1+\alpha/d$ [22, Thm. 1.2], or $q/p \le 1+\alpha/d$ [9, Thm. 1.1]; for *bounded* minimisers the condition is $q \le p+\alpha$ [23, 9]; and if $u \in C^{0,\gamma}$, $q < p+\alpha/(1-\gamma)$ suffices [9, Thm. 1.3]. For $a \in W^{1,\sigma}$, $\sigma > d$, De Filippis and Mingione obtain local Lipschitz bounds when $q/p \le 1 + 1/d - 1/\sigma$ (and, if $d = 2$, also $q/p < p$, as recorded in our source) [25, Thm. 1.3], and Hästö and Ok [42] prove $C^{1,\alpha}_{\mathrm{loc}}$ for general generalized Orlicz functions under their condition (VA1). All these results are *interior*.

We know of no refereed global (up to the boundary) gradient bound for x-dependent double phase problems; for these energies (S) must be *assumed*: it requires $u \in H_0^1$ and $|\nabla u|^{p-1} + a|\nabla u|^{q-1} \in L^2(\Omega)$ up to the boundary (automatic for the first term if $p \le 2$). Moreover, the two-phase experiments of Section 6 use an *indicator function* a, which is not Hölder continuous, so none of the above applies to them.

Energies not covered. For $G = e^t - 1 - t$, $tg'(t)/g(t) = te^t/(e^t-1) \to \infty$: the upper bound in (3.1) fails ($G \notin \Delta_2$), and so do the hypotheses $s_a < \infty$ of Facts 3.3 and 3.4. For $G = e^t - 1$ moreover $g(0) = 1 \neq 0$ and $tg'/g = t \to 0$ at 0. Interior Lipschitz bounds for exponential-type integrands appear in [25, Thms. 1.4–1.5], but we have not checked whether they cover these functions, and they are interior. Hence, for general data, *no result known to us gives (S) for* $e^t - 1 - t$ *or* $e^t - 1$, and we make no claim on the convergence of the continuous iteration for them; for the explicit radial problems of Section 6, (S) is verified directly (Remark 5.2).

3.3. **Failure of regularity: counterexamples.** The following examples show that the sufficient condition of Lemma 3.1 (a bounded gradient) can fail for non-uniformly elliptic or non-autonomous energies. We have not analysed whether (S) itself fails in them; in the presence of a Lavrentiev gap (Section 5), however, the conforming discrete problem is affected regardless of (S).

- *Esposito–Leonetti–Mingione* [32]: for integrands Hölder continuous in x with exponent α, regularity of minimisers in the sense of higher integrability ($Du \in L^q_{\mathrm{loc}}$; not $C^{1,\beta}$) holds when $q/p < (d+\alpha)/d$, and this bound is sharp: in the range $p < d < d+\alpha < q$ there are $a \in C^{0,\alpha}$ and boundary data with a Lavrentiev gap and a minimiser not in $W^{1,q}_{\mathrm{loc}}$ (extending Zhikov's two-dimensional construction).
- *Fonseca–Malý–Mingione* [34]: for $1 < p < d < d+\alpha < q$ and $\varepsilon > 0$ there is $a \in C^{0,\alpha}$ such that $\int(|Du|^p + a|Du|^q)$ has a local minimiser in $W^{1,p}$ whose singular set has Hausdorff dimension $> d - p - \varepsilon$; in particular it is discontinuous.
- *Zhikov* [52, 53]: checkerboard examples in two dimensions, for a variable exponent taking a value < 2 in two opposite quadrants and > 2 in the other two, and for the double phase energy with Lipschitz a and $p < 2 < 3 < q$, exhibit a Lavrentiev gap.
- *Balci–Diening–Surnachev* [6]: in $(-1,1)^d$, $d \ge 2$, for $p > 1$, $\alpha > 0$ and $q > p + \alpha\max\{1, (p-1)/(d-1)\}$ there is $a \in C^{0,\alpha}$ with a Lavrentiev gap [6, Thm. 32]; there are also variable exponents with modulus of continuity $1/\log(e + 1/t)^\kappa$, $\kappa < 1$ (just below log-Hölder), with a gap [6, Thm. 30]. (Theorem numbers refer to the arXiv version.)

TABLE 1. When is Assumption (S) known to hold? "via Lemma 3.1" means that the cited result gives a bounded gradient up to the boundary. Status of the sources is detailed in the text and in the file `TODO_VERIFY.md`.

Setting	(S) known?	Reference and hypotheses
Autonomous G, (3.1), $C^{1,\alpha}$ domain, $f \in L^\sigma$, $\sigma > d$	yes (via L. 3.1)	[3, Cor. 1.4]; [44, Thm. 1.7] (boundary part to be verified). x-dependence only as Hölder perturbation $\Lambda(1 + b(\lvert\xi\rvert))\lvert x - y\rvert^\alpha$ of a single b.
Autonomous G, Ω convex, $f \in L^{d,1}$	yes, conditionally (range of d unverified; $d \geq 3$ for [20]); not established for $d = 2$ from the sources read	[19] (abstract only), [20, Thm. 3.5] (p-type, $2 \leq p < d$, $d \geq 3$).
Autonomous G, $\Omega \subset \mathbb{R}^2$ convex, $f \in L^2$	yes, conditionally	[21, Thm. 2.3]: $a(\lvert\nabla u\rvert)\nabla u \in W^{1,2}$; notion of solution to be checked.
$p(x)$, $p \in C^{0,\mu}(\overline{\Omega})$, $\partial\Omega \in C^{1,\beta}$, bounded u	yes (via L. 3.1), conditionally	[33] (hypotheses not read in the original).
$p(x)$ log-Hölder only	unknown	Hölder continuity of u only [1].
$p(x)$ continuous, not log-Hölder	no bounded gradient in general	Lavrentiev gap possible [6, Thm. 30].
Double phase, $a \in C^{0,\alpha}$, $q/p \leq 1 + \alpha/d$; or bounded u, $q \leq p + \alpha$	interior only; must be assumed	[22, 23, 9]; [25] ($a \in W^{1,\sigma}$, $\sigma > d$); [42] (general).
Two-phase, a = indicator function	not covered	a not Hölder; no result applies.
Double phase, $q > p + \alpha \max\{1, \frac{p-1}{d-1}\}$, or $p < d < d + \alpha < q$	no bounded gradient in general; gap	[6, 32, 34, 53].
$e^t - 1 - t$, $e^t - 1$ (no Δ_2)	not available	outside (3.1); interior results of [25] not checked.
Explicit solutions (radial torsion, manufactured)	yes	Lemma 3.2, by direct verification.

- *Giaquinta, Marcellini, Hong* [38, 43], [48, §6]: unbounded minimisers for autonomous anisotropic integrands of (p, q)-growth (e.g. $p = 2$, $q = 4$, $d \geq 6$ in [38]; Marcellini's papers [47, 48] contain mainly positive results). These integrands are anisotropic; they do not concern isotropic energies $G(\lvert\xi\rvert)$ satisfying (3.1), for which Literature statement 3.3 gives bounded gradients. (For isotropic G violating (3.1), such as $e^t - 1 - t$, we know of no counterexample and no positive result; see the paragraph *Energies not covered.*)
- *Ball–Mizel* [8]: one-dimensional strictly convex, superlinear (but u-dependent) integrands whose minimiser is singular and does not satisfy the integrated Euler–Lagrange equation, with a Lavrentiev gap.

4. The discrete problem and convergence of the discrete iteration

4.1. **Discretisation.** Let $\Omega_h \subset \mathbb{R}^2$ be a polygon (for the disk, the inscribed polygon whose vertices are the boundary nodes) with a conforming triangulation $\mathcal{T}_h$, x_K the barycentre of $K \in \mathcal{T}_h$, and

$$V_h = \{v \in C(\overline{\Omega_h}) : v|_K \in \mathbb{P}_1 \ \forall K,\ v = 0 \text{ on } \partial\Omega_h\}, \qquad H_h = \{q\colon \Omega_h \to \mathbb{R}^2 :\ q|_K \text{ constant } \forall K\},$$

with the $L^2(\Omega_h)$ inner product $(q, q')_h = \sum_K |K|\, q_K \cdot q'_K$ on H_h, the norm $\|v\|_{V_h} = \|\nabla v\|_h$ on V_h, and $B_h = \nabla\colon V_h \to H_h$. For the discretisation we assume that $G(x,\cdot)$ is a convex Young function for *every* $x \in \overline{\Omega}_h$ (as in all our examples) and that f has well-defined values at the barycentres (e.g. $f \in C(\overline{\Omega_h})$, or f piecewise continuous with the chosen values at its discontinuities); otherwise F_h and ℓ_h are to be defined with a fixed representative, or ℓ_h by exact integration. The Young function is sampled at the barycentres and the load is integrated by the barycentre rule:

$$F_h(q) = \sum_K |K|\, G(x_K, |q_K|), \qquad \ell_h(w) = \sum_K |K|\, f(x_K)\, w(x_K), \qquad J_h(v) = F_h(\nabla v) - \ell_h(v).$$

(Any other linear functional ℓ_h on V_h may be used; the code also offers a three-point rule.) $\|\cdot\|_{V_h}$ is a norm: if $\nabla v = 0$ then v is constant on each connected component of Ω_h and vanishes on its boundary. Thus $(V_h, H_h, B_h, F_h, \ell_h)$ satisfies (A1)–(A3), with F_h finite, convex and continuous.

The algorithm (2.3)–(2.5) for these data is exactly what the code computes: with A the P1 stiffness matrix restricted to the free nodes, step (2.3) is the linear system $rAU_n = b_h + \big[(r\nu_{n-1} - \eta_n, \nabla\phi_i)_h\big]_i$, where b_h represents ℓ_h; A is factorised once. Step (2.4) decouples into one problem per triangle, $\min_{\xi\in\mathbb{R}^2} G(x_K, |\xi|) - \eta_{n,K}\cdot\xi + \frac{r}{2}|\nabla u_n|_K - \xi|^2$, solved by Lemma 2.7: a bisection on the scalar equation (2.8) in $[0, |w_K|]$ (in the code, in the variable $\log t$, so that very small roots are resolved to relative accuracy). Step (2.5) is explicit, and $\eta_{n+1}|_K \in \partial\Phi(x_K,\cdot)(\nu_n|_K)$ is the discrete flux.

4.2. **Theorem C.**

Theorem C. *Let G be as in Section 4.1 ($G(x,\cdot)$ a convex Young function for every x) and such that $G(x_K,\cdot)$ is superlinear for every $K \in \mathcal{T}_h$:*

$$\lim_{t\to\infty} G(x_K, t)/t = +\infty. \tag{4.1}$$

Then:

(i) *J_h has a minimiser $u_h \in V_h$; it is unique if each $G(x_K,\cdot)$ is strictly convex.*

(ii) *There is $\eta_h \in H_h$ such that $(u_h, \nabla u_h, \eta_h)$ is a saddle point of the discrete augmented Lagrangian, for every $r > 0$; that is, $\eta_h|_K \in \partial\Phi(x_K,\cdot)(\nabla u_h|_K)$ for all K and $(\eta_h, \nabla w)_h = \ell_h(w)$ for all $w \in V_h$.*

(iii) *For every $r > 0$, every $\nu_0 \in H_h$ and every $\eta_1 \in \partial F_h(\nu_0)$ (e.g. $\nu_0 = \eta_1 = 0$), the iterates converge: $u_n \to \bar{u}_h$, $\nu_n \to \nabla\bar{u}_h$, $\eta_n \to \bar{\eta}_h$, where $(\bar{u}_h, \nabla\bar{u}_h, \bar{\eta}_h)$ is a discrete saddle point; $\bar{u}_h$ minimises J_h, and $\nabla u_n - \nu_n \to 0$. The identity (2.9) holds with $(\cdot,\cdot)_h$, and E_n is nonincreasing.*

This applies to t^p/p for all $p > 1$, to $p(x)$, p-q and two-phase energies (with $p > 1$), to $t^p \log^a(1+t)$ whenever it is a convex Young function with $p > 1$, or $p = 1$ and $a > 0$, to $e^t - 1 - t$ and to $e^t - 1$, with no regularity assumption and no restriction on r. Superlinearity cannot be dropped: for $G \equiv 0$, or for $G(t) = t$ and ℓ_h large, J_h is unbounded below and no saddle point exists.

Proof. (i) Let $L = \sup\{|\ell_h(v)| : \|\nabla v\|_h \le 1\} < \infty$ and $a_{\min} = \min_K |K|$. For $q \in H_h$, $\|q\|_h^2 = \sum_K |K||q_K|^2 \le \max_K |q_K| \sum_K |K||q_K| \le a_{\min}^{-1}\big(\sum_K |K||q_K|\big)^2$, i.e. $\|q\|_{L^1} \ge a_{\min}^{1/2}\|q\|_h$. Given $M > 0$, by (4.1) there is t_M with $G(x_K, t) \ge Mt$ for $t \ge t_M$ and all (finitely many) K; since $G \ge 0$, $G(x_K, t) \ge Mt - Mt_M$ for all $t \ge 0$. Hence

$$J_h(v) \ge M\|\nabla v\|_{L^1} - Mt_M|\Omega_h| - L\|\nabla v\|_h \ge \big(Ma_{\min}^{1/2} - L\big)\|\nabla v\|_h - Mt_M|\Omega_h|,$$

and choosing $Ma_{\min}^{1/2} > L$ shows that J_h is coercive on the finite-dimensional space V_h. J_h is continuous (each $G(x_K,\cdot)$ is a finite convex function, hence continuous), so it attains its minimum on a nonempty compact sublevel set. If each $\gamma = G(x_K,\cdot)$ is strictly convex, then $\phi = \Phi(x_K,\cdot)$ is strictly convex: γ is strictly increasing (if $\gamma(a) = \gamma(b)$ with $a < b$, strict convexity would give $\gamma((a+b)/2) < \gamma(a)$, against monotonicity); for $\xi_1 \neq \xi_2$ with $|\xi_1| = |\xi_2|$ we have $|(\xi_1+\xi_2)/2| < |\xi_1|$, hence $\phi((\xi_1+\xi_2)/2) < \phi(\xi_1) = \frac{1}{2}(\phi(\xi_1)+\phi(\xi_2))$, and for $|\xi_1| \neq |\xi_2|$, $\phi((\xi_1+\xi_2)/2) \leq \gamma(\frac{|\xi_1|+|\xi_2|}{2}) < \frac{1}{2}(\phi(\xi_1)+\phi(\xi_2))$; so F_h and, ∇ being injective, J_h are strictly convex.

(ii) Apply Lemma 2.14 with $X = H_h$, $R = \nabla(V_h)$, $\lambda \in R$ with $(\lambda, \nabla w)_h = \ell_h(w)$ for $w \in V_h$, $\Psi(q) = F_h(q) - (\lambda, q)_h$ and $q_0 = \nabla u_h$. Ψ is finite and convex on the finite-dimensional space H_h, hence bounded above on a neighbourhood of q_0 (on a simplex containing q_0 in its interior it is bounded by its values at the vertices). Since $\Psi(\nabla v) = J_h(v)$ for $v \in V_h$, q_0 minimises Ψ on R. The lemma gives $\zeta \in \partial\Psi(q_0)$, $\zeta \perp R$; then $\eta_h = \zeta + \lambda \in \partial F_h(\nabla u_h)$ and $(\eta_h, \nabla w)_h = \ell_h(w)$. By Proposition 2.3 (which uses only (A1)–(A3)) $(u_h, \nabla u_h, \eta_h)$ is a saddle point, for every r. The pointwise form of ∂F_h is the finite-dimensional analogue of Lemma 2.4 (F_h is a sum of functions of the separate blocks q_K, and $|K| > 0$).

(iii) By (ii), Assumption (S) holds for the discrete data, and Theorem A, whose proof uses only (A1)–(A3) (Rellich's theorem is only used for the concrete L^2 statement), gives weak convergence of (u_n, ν_n, η_n) to a saddle point, $\nabla u_n - \nu_n \to 0$, and (2.9). In finite dimension weak and strong convergence coincide. By Proposition 2.3, $\bar{u}_h$ minimises J_h. □

Remark 4.1 (Nonhomogeneous Dirichlet data). For $u = u_D$ on $\partial\Omega_h$, let $\tilde{u}_h$ be the P1 function with the prescribed boundary values and zero interior values. Writing $u = \tilde{u}_h + v$, $\nu = \nabla\tilde{u}_h + q$, the iteration of the code in the variables (u, ν, η) is exactly the algorithm (2.3)–(2.5) in the variables (v, q, η) for $\tilde{F}_h(q) = F_h(q + \nabla\tilde{u}_h)$, which is again finite, convex and superlinear, and nonnegative, but with $\tilde{F}_h(0) = F_h(\nabla\tilde{u}_h) \neq 0$ in general; this is why (A2) only asks F to be proper. The initialisation $\nu_0 = \eta_1 = 0$ of the code corresponds to $q_0 = -\nabla\tilde{u}_h$, $\eta_1 = 0 \in \partial\tilde{F}_h(q_0) = \partial F_h(0)$, which is admissible (and by Remark 2.9 any initialisation is). The existence of a discrete minimiser and of the multiplier (Theorem C(i)–(ii)) follows with the same proof. Hence Theorem C also covers the manufactured test of Section 6.4.

Remark 4.2 (Nonhomogeneous data at the continuous level). If $u = u_D$ on $\partial\Omega$ with $u_D \in H^1(\Omega)$ and $F(\nabla u_D) < \infty$, write $u = u_D + v$, $v \in H_0^1(\Omega)$, and $\tilde{F}(q) = F(q + \nabla u_D)$, $\tilde{\ell}(v) = (f, v)$. Then $(V, H, \nabla, \tilde{F}, \tilde{\ell})$ satisfies (A1)–(A3), and Proposition 2.3 and Theorems A–B apply in the variables (v, q, η), with $\nu = q + \nabla u_D$; the proofs of Theorem B use only $\partial\tilde{F}(q) = \partial F(q + \nabla u_D) = \{T(\cdot, q + \nabla u_D)\}$ and the differences $\nu_n - \nu^* = q_n - q^*$, so they go through unchanged. Lemma 3.2 holds with "$u \in H_0^1$" replaced by "$u - u_D \in H_0^1$".

Remark 4.3 (Related results and what is not claimed). Theorem C(iii) can also be obtained from finite-dimensional ADMM theory, e.g. [31, Thm. 8] (which requires a Kuhn–Tucker pair and full column rank of the constraint matrix, here ∇) or, for the residual and the dual variables, [14, §3.2]; the point of (i)–(ii) is that the saddle point always exists. For the $p(x)$-Laplacian discretised by discontinuous Galerkin methods, Del Pezzo and Martínez [26] proved convergence of ALG1/ALG2 by applying Glowinski's theorems in finite dimension, where F is finite, differentiable and superlinear; their injectivity hypothesis fails for their lifted gradient, and they use it only for the convergence of u_n. Theorem C says nothing about the *speed* of convergence or its dependence on h, p or r: in the experiments the number of iterations grows as h decreases for $p < 2$, $p = 4$, $t^2\log(1+t)$ and $e^t - 1$ (Table 4), and for ADMM on finite element discretisations Bartels and Milicevic observe h-dependent constants [11]. Nor does it say anything about the convergence of u_h to the solution of (1.1) as $h \to 0$; see Section 5 and Section 6.8.

5. The Lavrentiev phenomenon and the discrete problem

Let $\mathcal{W}$ denote the finite-energy functions in $W^{1,1}_0(\Omega)$ (with the given boundary data) and consider $m_{\mathcal{W}} = \inf_{\mathcal{W}} J$ and the infimum $m_{\mathcal{H}}$ of J over smooth functions with the same boundary data. A *Lavrentiev gap* is the situation $m_{\mathcal{H}} > m_{\mathcal{W}}$. (Below we also use the infimum $m_{\mathrm{Lip}} \geq m_{\mathcal{W}}$ over Lipschitz functions; conforming P1 functions are Lipschitz.) A gap it is the variational counterpart of the non-density of smooth functions in the energy space ($H \neq W$ in Zhikov's notation [52, 53]). What is known for the energies of interest here:

- *Gap.* Double phase with $a \in C^{0,\alpha}$ and $q > p + \alpha \max\{1, (p-1)/(d-1)\}$ [6, Thm. 32]; earlier examples in the range $p < d < d+\alpha < q$ [53, 32]; variable exponents with modulus of continuity just below log-Hölder [6, Thm. 30], and Zhikov's discontinuous checkerboard exponent [52].
- *No gap.* On bounded Lipschitz domains with $a \in C^{0,\alpha}(\overline{\Omega})$, boundary data in $W^{1,q}$ and $q \leq p + \alpha \max\{1, p/d\}$, the infima over $W^{1,p}$, $W^{1,q}$ and C^∞_c perturbations coincide, without any boundedness assumption [17, Thm. 2.3]; this range is sharp for $p \leq d$ by [6]. Local versions under boundedness or Hölder continuity of u are in [9, Thm. 1.4]; see also the sharp scale of weights in [13]. For variable exponents, log-Hölder continuity excludes the gap (attributed to Zhikov, cf. [53]).

Consequence for conforming discretisations. Conforming P1 functions are Lipschitz. Carstensen and Ortner [18] show that conforming finite element methods fail exactly when the Lavrentiev phenomenon is present, and Balci, Ortner and Storn [7] explain that conforming discrete minima converge (with exact or sufficiently accurate quadrature) to the infimum over the closure of smooth functions, not to $m_{\mathcal{W}}$, and observe the resulting energy gap in two-dimensional double phase and variable exponent examples; nonconforming Crouzeix–Raviart elements converge to the true minimum [7, Thm. 1], see also [50, 5]. Therefore, in the presence of a gap, *the discrete problem solved by our method need not approximate the continuous minimiser.* Theorem C is not affected: the iteration still converges, for every r, to the discrete minimiser u_h; what fails is the approximation property of u_h as $h \to 0$. Note also that our discretisation samples $G(x,\cdot)$ at barycentres, which is a quadrature, and [7] report that conforming schemes are sensitive to quadrature in the gap regime.

Our test problems. For the test problems with an exact solution in Section 6 we can say precisely what holds. The following elementary lemma gives a lower bound for every conforming function; it does *not* by itself give convergence of u_h to u.

Lemma 5.1. *Let $f \in L^\infty(\Omega)$, $u \in W^{1,\infty}(\Omega)$ with $u - u_D \in W^{1,1}_0(\Omega)$, and $\eta \in W^{1,\infty}(\Omega;\mathbb{R}^d)$ with $-\operatorname{div}\eta = f$ and $\eta(x) \in \partial\Phi(x,\cdot)(\nabla u(x))$ a.e. Then $J(v) \geq J(u)$ for every $v \in u_D + W^{1,1}_0(\Omega)$ with $J(v)$ defined; in particular u is a Lipschitz minimiser over the whole finite-energy class, and no function, conforming finite element functions included, has smaller energy.*

Proof. The pointwise subgradient inequality gives $\Phi(x,\nabla v) \geq \Phi(x,\nabla u) + \eta\cdot\nabla(v-u)$, and $\int \eta\cdot\nabla w = -\int \operatorname{div}\eta\, w = \int fw$ for $w \in W^{1,1}_0$ (by density of C^∞_c in $W^{1,1}_0$, both sides being continuous in $W^{1,1}$ since $\eta, \operatorname{div}\eta \in L^\infty$). Integrating with $w = v - u$ gives $J(v) \geq J(u)$. Note that $J(u)$ is finite: the subgradient inequality at $\xi = 0$ gives $0 \leq \Phi(x,\nabla u) \leq \eta\cdot\nabla u \in L^\infty$. □

Thus $m_{\mathcal{W}} = m_{\mathrm{Lip}} = J(u)$ (no gap between $W^{1,1}$ and Lipschitz functions), and $J(v_h) \geq J(u)$ for every conforming finite element function v_h.

Remark 5.2 (The radial and manufactured test problems). Consider the unit disk, $f = c > 0$, and a radial Young integrand $G(x,t) = G_0(|x|,t)$ such that, as in all our examples, $g_0(s,\cdot) = \partial_t G_0(s,\cdot)$ is continuous and strictly increasing on $(0,\infty)$ and $\inf_s g_0(s,t) \to \infty$ as $t \to \infty$ (in our examples $g_0(s,t) \geq g_{\min}(t)$ with $g_{\min}(t) = t^{p-1}$, $t^{1/2}$, t, or $e^t - 1$). Let $\tau(s) = 0$ if $cs/2 \leq g_0(s,0)$ and otherwise let $\tau(s) > 0$ solve $g_0(s,\tau(s)) = cs/2$; τ is measurable, being a monotone inverse ($\tau(s) = \sup\{t \in \mathbb{Q}_{\geq 0} : g_0(s,t) < cs/2\}$), and bounded, since $\tau(s) \leq t^*$ whenever $\inf_s g_0(s,t^*) \geq c/2$. The solution is $u(x) = \int_{|x|}^1 \tau(s)\,ds$ and the flux is $\eta(x) = -cx/2$. Indeed $|\eta| = c|x|/2$, η points in the

direction of $\nabla u = -\tau(|x|)x/|x|$, so $\eta(x) \in \partial\Phi(x,\cdot)(\nabla u(x))$ by Lemma 2.5 (part (i) in the dead core of $e^t - 1$, where $\nabla u = 0$ and $|\eta| \le 1 = g(0)$; part (ii) elsewhere, the interface $|x| = 1/2$ of the two-phase case being a null set), and $-\operatorname{div}\eta = c$. Since τ is bounded, u is Lipschitz and vanishes on ∂B_1. By Lemmas 3.2 and 5.1, *for these particular data* (S) holds and there is no Lavrentiev obstruction, also for $e^t - 1 - t$, for $e^t - 1$, and for the two-phase coefficient $a = \mathbf{1}_{\{|x|>1/2\}}$, which are outside the regularity theories of Section 3 and outside the Hölder hypothesis of the no-gap results above. Consequently Theorem A applies to the *continuous* iteration for every radial test problem, and Theorem B applies as indicated in Table 3 (the computed, discrete iteration is covered by Theorem C in all cases). The same argument applies to the manufactured solution of Section 6.4 ($u = e^{x_1+x_2}$ and its flux are smooth on the closed square), with the nonhomogeneous data handled by Remark 4.2. This is a statement about these explicit data only. For the two-phase problem on a sector (Section 6.7) no exact solution is known, a is an indicator function, and we can exclude neither the failure of (S) nor a Lavrentiev gap by any result we know; that experiment is qualitative.

For the autonomous radial problems the lower bound of Lemma 5.1 can be combined with interpolation to prove convergence of the exact discrete minimisers.

Proposition 5.3. *Let $G = G(t)$ be independent of x and strictly convex, with g continuous and strictly increasing on $(0,\infty)$ and $g(t) \to \infty$; let $\Omega = B_1 \subset \mathbb{R}^2$, $f = c > 0$, and let u, η be as in Remark 5.2. Let $(\mathcal{T}_h)$ be a shape-regular family of triangulations of polygons $\Omega_h \subset B_1$ whose boundary vertices lie on ∂B_1, with maximal element diameter $h \to 0$, and let u_h be the minimiser of J_h (Theorem C), extended by 0 to B_1. Then:*

(i) *$J(u_h) \to J(u)$, and $D_h := \int_{B_1} \big[\Phi(\nabla u_h) - \Phi(\nabla u) - \eta \cdot \nabla(u_h - u)\big]\,dx = J(u_h) - J(u) \to 0$;*
(ii) *$\nabla u_h \to \nabla u$ in measure in B_1;*
(iii) *if moreover $g(0) = 0$ and $k(t) = g(t)/t$ is nondecreasing and positive on $(0,\infty)$, then $u_h \to u$ in $H^1_0(B_1)$.*

Proof. For $v \in V_h$ extended by 0, $J_h(v) = J(v)$: G does not depend on x and ∇v is piecewise constant, so $F_h(\nabla v) = \int_{B_1} G(|\nabla v|)$ (recall $G(0) = 0$), and the barycentre rule is exact for $\int_K cv$ because $v|_K$ is affine. Moreover $v \in W^{1,\infty}_0(B_1)$.

(i) By Lemma 5.1, $J(u_h) \ge J(u)$, and integrating the definition of D_h with $\int \eta \cdot \nabla w = \int cw$ ($w \in W^{1,1}_0$) gives $D_h = J(u_h) - J(u) \ge 0$. The nodal interpolant $I_h u$ belongs to V_h because $u = 0$ on ∂B_1, so $J(u_h) = J_h(u_h) \le J_h(I_h u) = J(I_h u)$. Let $L = \sup\tau$ be the Lipschitz constant of u. On each triangle, $\nabla I_h u$ is determined by the differences of u along two edges, each bounded by L times the edge length, so $|\nabla I_h u| \le C_\sigma L$ with C_σ depending only on the shape regularity. Since g is continuous and strictly increasing, τ is continuous on $[0,1]$ and $\nabla u(x) = -\tau(|x|)x/|x|$ is continuous on $\overline{B_1} \setminus \{0\}$; by the mean value theorem along the edges, $|\nabla I_h u|_K - \nabla u(x)| \le C_\sigma \operatorname{osc}_K \nabla u$ for $x \in K$, which tends to 0 for every $x \ne 0$ in B_1 (for h small, $x \in \Omega_h$, because the boundary edges have length $\le h$). Also $|B_1 \setminus \Omega_h| \to 0$ and $|I_h u| \le \sup|u|$. By dominated convergence (majorant $G(C_\sigma L) + c\sup|u|$), $J(I_h u) \to J(u)$. Hence $J(u) \le J(u_h) \le J(I_h u) \to J(u)$.

(ii) Let $\varepsilon > 0$ and $Q = \{(a,\zeta) : |a| \le L,\ |\zeta| \le c/2,\ \zeta \in \partial\Phi(a)\}$, a compact set (the graph of $\partial\Phi$ is closed, Φ being continuous and convex), which contains $(\nabla u(x), \eta(x))$ for a.e. x. For $(a,\zeta) \in Q$ the convex function $D_{a,\zeta}(b) = \Phi(b) - \Phi(a) - \zeta \cdot (b - a)$ is ≥ 0 and vanishes at a; hence it is nondecreasing along rays from a, and $D_{a,\zeta}(b) \ge \mu(\varepsilon) := \min\{D_{a,\zeta}(z) : (a,\zeta) \in Q,\ |z - a| = \varepsilon\}$ when $|b - a| \ge \varepsilon$. The minimum is attained and positive: if $D_{a,\zeta}(z) = 0$ with $z \ne a$, then $\Phi(\frac{a+z}{2}) \ge \Phi(a) + \zeta \cdot \frac{z-a}{2} = \frac{1}{2}(\Phi(a) + \Phi(z))$, contradicting the strict convexity of Φ (proved in Theorem C(i)). Therefore $|\{|\nabla u_h - \nabla u| \ge \varepsilon\}| \le D_h/\mu(\varepsilon) \to 0$.

(iii) Now $\Phi \in C^1$ and $\eta = T(\nabla u)$ (Lemma 2.5(iii)). With $a = \nabla u(x)$, $b = \nabla u_h(x)$, and ψ as in Section 2.5,

$$\begin{aligned}\Phi(b) - \Phi(a) - T(a)\cdot(b-a) &= \int_0^1 \big(T(a+s(b-a)) - T(a)\big)\cdot(b-a)\,ds \\ &= \int_0^1 \psi\big(a+s(b-a),a\big)\frac{ds}{s} \geq \frac{1}{8}\,k\Big(\frac{|b-a|}{4}\Big)|b-a|^2,\end{aligned}$$

by Lemma 2.12(i) and the monotonicity of k, restricting the integral to $s \in [\frac{1}{2},1]$. Splitting B_1 into $\{|\nabla u_h - \nabla u| \leq \delta\}$ and its complement as in the proof of Theorem B(a) gives $\|\nabla(u_h - u)\|^2 \leq \delta^2|B_1| + 8D_h/k(\delta/4)$, so $\nabla u_h \to \nabla u$ in $L^2(B_1)$. □

Proposition 5.3 applies to all autonomous radial test problems of Table 3 (parts (i)–(ii)) and, for part (iii), to $t^2/2$, $t^4/4$, $t^2/2 + t^4/4$, $t^2\log(1+t)$, $t^3/\log(1+t)$ and $e^t - 1 - t$. It concerns the exact discrete minimisers u_h, not the computed iterates, and gives no rate. It does not cover the two-phase problem with $a = \mathbf{1}_{\{|x|>1/2\}}$: there $J_h \neq J$ on the triangles cut by the interface, and controlling this consistency error would require a uniform bound on ∇u_h, which we do not have; for that problem, and in general, we do not prove that $u_h \to u$.

6. Numerical experiments

6.1. **Protocol.** All computations use the package `glaplace` (P1 elements, piecewise constant ν, η, G sampled at barycentres, one sparse LU factorisation per problem, vectorised bisection for (2.8)). The disk is meshed by a DistMesh-type generator with target edge lengths $h = 0.2$, 0.1, 0.05, 0.025 and 0.0125 (from 88 to 23 227 nodes; the actual maximal edge lengths are 0.288, 0.136, 0.0625, 0.0313 and 0.0161); the meshes are unstructured and Ω_h is the inscribed polygon. The iteration starts from $\nu_0 = \eta_1 = 0$ and stops when the scaled primal residual

$$\|\nabla u_n - \nu_n\|_h/(1 + \|\nabla u_n\|_h) < \mathrm{tol}$$

(which tends to 0 by Theorem C), or after a maximum number of iterations. Errors are *true* finite element errors: $\|u - u_h\|_{L^2(\Omega_h)}$ and $\|\nabla(u - u_h)\|_{L^2(\Omega_h)}$ are computed with a six-point degree-4 Dunavant rule on every triangle, relative to the corresponding norm of u; the maximum error is taken at the nodes (absolute). Orders κ in error $\approx Ch^\kappa$ are least-squares fits (in the target h) over the three finest meshes. The exact radial solutions $u(x) = \int_{|x|}^1 g^{-1}(cs/2)\,ds$ are computed by bisection for g^{-1} and composite Gauss–Legendre quadrature; we checked them against the closed forms for t^p/p ($p = 1.5, 2, 4$; maximal relative deviation 9×10^{-9} for $p = 4$, and $\leq 3 \times 10^{-15}$ for $p = 1.5, 2$; file `results/exactcheck.txt`); for the other G we did not check them independently. All numbers below were produced by the scripts listed in `NOTES.md`; timings are not reported, since they were measured in a shared environment.

Why not nodal differences. The L^2 norm of the P1 function $I_h u - u_h$ (nodal difference) is not the error: $I_h u$ is superclose to u_h. Table 2 shows that it underestimates the true L^2 error by factors between 7 and 25 on the finest mesh and suggests spurious orders between 2.1 and 2.5. We report only true errors.

6.2. **Torsion on the disk with exact solutions.** We solve (1.2) on the unit disk with $f = c$ and $u = 0$ on the boundary, for the ten Young functions of Table 3: $c = 1$, $r = 1$, tolerance 10^{-9}, at most 3000 iterations, except for $e^t - 1$ ($c = 8$, $r = 30$), whose solution has a dead core $\{|x| \leq 1/4\}$ where $\nabla u = 0$. The two-phase case uses $G(x,t) = t^2/2 + a(x)t^4/4$ with $a = \mathbf{1}_{\{|x|>1/2\}}$, an interface not aligned with the mesh. Figure 1 shows the errors and Tables 3–4 the fitted orders, the errors on the finest mesh and the iteration counts.

TABLE 2. True relative L^2 error versus the nodal quantity $\|I_h u - u_h\|_{L^2}/\|I_h u\|_{L^2}$, radial torsion, $c = 1$, $r = 1$, tolerance 10^{-10}. Orders fitted over $h = 0.05, 0.025, 0.0125$.

	$h = 0.0125$		fitted order		ratio
G	true	nodal	true	nodal	true/nodal
$t^2/2$	7.04×10^{-5}	2.79×10^{-6}	2.00	2.53	25.2
$t^4/4$	6.13×10^{-5}	8.89×10^{-6}	2.00	2.08	6.9
$t^2/2 + t^4/4$	6.46×10^{-5}	3.97×10^{-6}	2.00	2.44	16.3
$e^t - 1 - t$	6.57×10^{-5}	3.63×10^{-6}	2.00	2.46	18.1

TABLE 3. Radial torsion on the unit disk: fitted orders (three finest meshes) of the relative L^2 errors of u and ∇u and of the nodal maximum error; errors on the finest mesh ($h = 0.0125$, 23 227 nodes). Column "theory": which result applies to the *continuous* iteration, given that (S) holds for these data (Remark 5.2); the discrete iteration is covered by Theorem C in all cases. "L" marks the functions satisfying (3.1). †: two of the three meshes used in the fit did not reach the tolerance (Table 4).

case	G	theory	order			error at $h = 0.0125$	
			$L^2(u)$	$L^2(\nabla u)$	max	$L^2(u)$	$L^2(\nabla u)$
p1.5	$t^{1.5}/1.5$	B(b), L	2.01	1.02	1.94	1.15×10^{-4}	7.47×10^{-3}
p2	$t^2/2$	B(a), L	2.00	1.02	2.02	7.04×10^{-5}	5.14×10^{-3}
p4	$t^4/4$	B(a), L	2.00	1.00	1.17	6.13×10^{-5}	5.85×10^{-3}
pq24	$t^2/2 + t^4/4$	B(a), L	2.00	1.01	2.02	6.46×10^{-5}	4.94×10^{-3}
pq1.5_3	$t^{1.5}/1.5 + t^3/3$	A, L	2.01	1.02	1.98	9.73×10^{-5}	6.69×10^{-3}
plog2_1	$t^2 \log(1+t)$	B(a), L	2.00	1.01	1.37	6.00×10^{-5}	5.11×10^{-3}
plog3_-1	$t^3/\log(1+t)$	B(a), L	2.00	1.02	2.02	6.72×10^{-5}	5.02×10^{-3}
exp	$e^t - 1 - t$	B(a)	2.00	1.01	2.02	6.57×10^{-5}	4.97×10^{-3}
twophase	$t^2/2 + \mathbf{1}_{\lvert x\rvert > \frac{1}{2}} t^4/4$	B(a)	1.90	0.98	0.79	7.26×10^{-5}	5.29×10^{-3}
expdead†	$e^t - 1$, $c = 8$	A	1.97	1.01	1.11	7.25×10^{-5}	5.66×10^{-3}

Observations. (1) For all cases with smooth G the observed orders are 2 for $\|u - u_h\|_{L^2}$ and 1 for $\|\nabla(u - u_h)\|_{L^2}$, independently of G, including $e^t - 1 - t$ (no Δ_2), $t^3/\log(1+t)$ and the sub-quadratic $t^{1.5}/1.5$. (2) The nodal maximum error has order about 2 for most cases but 1.17 for $t^4/4$, 1.37 for $t^2\log(1+t)$ and 1.11 for $e^t - 1$ (two-phase: see (3)); the pairwise orders fluctuate (for $t^4/4$: 0.91, 1.55, 1.81, 0.52). We attribute this, without proof, to the limited smoothness of u at the origin, where $\nabla u = 0$ and g^{-1} is not smooth ($|\nabla u| \sim |x|^{1/3}$ for $p = 4$). (3) For the two-phase coefficient, whose jump at $|x| = 1/2$ is not resolved by the mesh and is sampled at barycentres, the orders drop to 1.90 (L^2) and 0.79 (maximum), consistently with the jump of u' across the interface. (4) For $e^t - 1$, the iteration is slow (as for the 1-Laplacian, $g(0) > 0$ makes ∂F multivalued); on the two finest meshes the tolerance was not reached in 3000 iterations (residuals 1.9×10^{-8} and 3.9×10^{-8}), but the errors are in line with the coarser meshes (order 1.97 in L^2, from a fit that includes these two unconverged runs). The computed solution is flat in the dead core: on the mesh $h = 0.05$ the discrete gradient satisfies $\max|\nabla u_h| \le 1.6 \times 10^{-7}$ on the triangles with barycentre in $\{|x| < 0.2\}$ (Table 7, Figure 2). (5) The number of iterations is essentially independent of h for $t^2/2$, $t^2/2 + t^4/4$, $e^t - 1 - t$ and the two-phase case, and grows as h decreases for $t^{1.5}/1.5$, $t^{1.5}/1.5 + t^3/3$, $t^4/4$, $t^2\log(1+t)$ and $e^t - 1$ (Table 4); for $t^{1.5}/1.5$ the successive ratios correspond to exponents 0.94,

TABLE 4. Iterations to reach the tolerance 10^{-9} for the cases of Table 3 ($r = 1$; $r = 30$ for expdead). Entries 3000* did not reach the tolerance (final residuals 1.9×10^{-8} and 3.9×10^{-8}).

h	0.2	0.1	0.05	0.025	0.0125
nodes	88	362	1452	5809	23227
p1.5	147	282	439	584	837
p2	29	29	29	29	29
p4	47	73	109	123	176
pq24	30	29	28	27	26
pq1.5_3	148	282	441	587	843
plog2_1	36	37	45	50	60
plog3_-1	41	39	37	35	33
exp	29	28	27	26	25
twophase	30	29	28	27	27
expdead	252	261	327	3000*	3000*

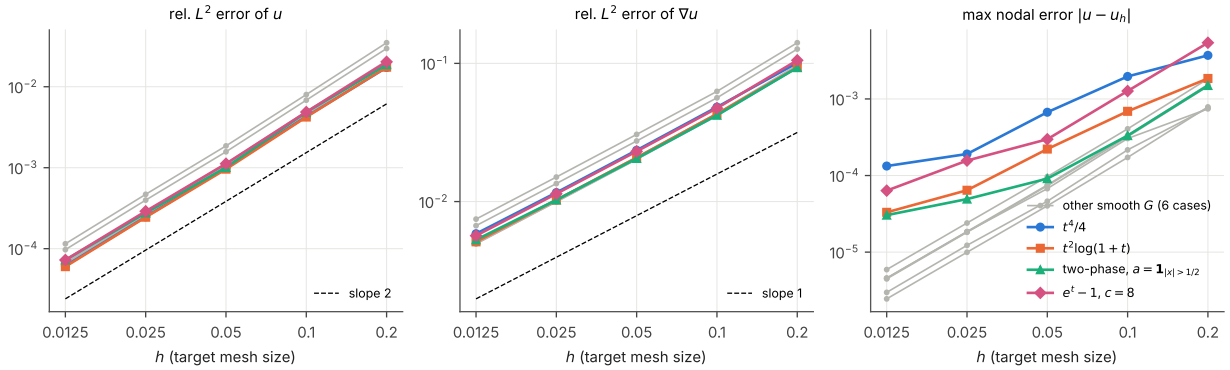


FIGURE 1. Radial torsion: relative L^2 errors of u (left) and ∇u (centre), and nodal maximum error (right), versus the target mesh size. Gray: the six cases $t^{1.5}/1.5$, $t^2/2$, $t^2/2 + t^4/4$, $t^{1.5}/1.5 + t^3/3$, $t^3/\log(1+t)$, $e^t - 1 - t$; the four cases with reduced maximum-norm order are highlighted. Raw data: `convergence_order.csv`.

0.64, 0.41, 0.52 of h^{-1}, i.e. roughly $h^{-1/2}$ on the finest meshes; Theorem C gives no information on this.

6.3. **The p-Laplacian for p close to 1.** For t^p/p with $p = 1.2$ and $p = 1.1$ ($c = 1$, $r = 1$, tolerance 10^{-9}, at most 6000 iterations, $h = 0.2, \ldots, 0.025$), *none* of the eight runs reached the tolerance; the final residuals were between 1.15×10^{-7} and 4.1×10^{-7} for $p = 1.2$ and between 6.3×10^{-7} and 3.9×10^{-6} for $p = 1.1$. The errors are reported with this caveat (Table 5); since a small residual does not control the iteration error (Section 6.5), we recorded the L^2 error during the iteration on the two finest meshes. For $p = 1.2$ it changes by less than 3×10^{-5} in relative terms between iterations 3000 and 6000 (6.2583×10^{-3} vs 6.2584×10^{-3} for $h = 0.05$, 1.59667×10^{-3} vs 1.59672×10^{-3} for $h = 0.025$), so the iteration error is negligible there. For $p = 1.1$ it is still drifting: 2.124×10^{-2} vs 2.182×10^{-2} ($h = 0.05$) and 5.814×10^{-3} vs 5.960×10^{-3} ($h = 0.025$), i.e. by about 3%, so the $p = 1.1$ errors contain an iteration component of a few percent (file `results/pto1_hist.txt`). The fitted L^2 orders are 1.98 ($p = 1.2$) and 1.84 ($p = 1.1$), and the gradient orders are 1.03 and 1.02; all fits use only unconverged runs. The error constants deteriorate as $p \to 1$: on the mesh $h = 0.025$ the relative L^2 error is 1.6×10^{-3} for $p = 1.2$ and 6.0×10^{-3} for $p = 1.1$, against 4.7×10^{-4} for

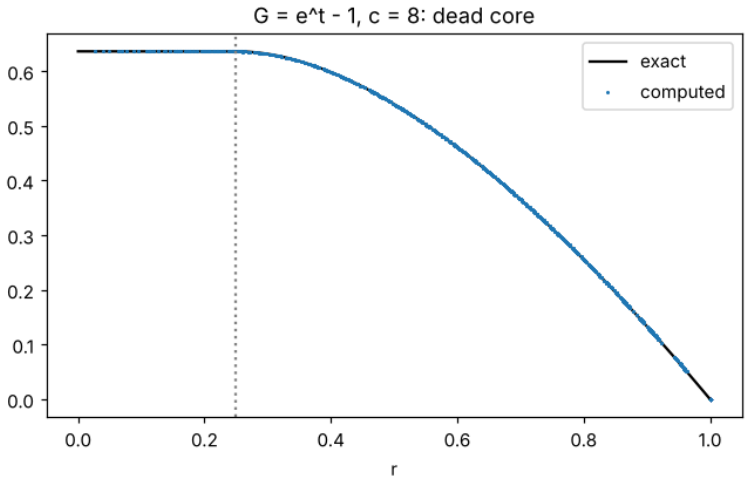


FIGURE 2. $G = e^t - 1$, $c = 8$: exact radial solution (line) and nodal values of u_h ($h = 0.05$, $r = 30$) against $|x|$; the dotted line marks the boundary $|x| = 2/c = 1/4$ of the dead core.

TABLE 5. t^p/p, p close to 1: relative true errors after 6000 iterations (no run reached the tolerance 10^{-9}; final residual in the last column). The maximum of the exact solution is 5.2×10^{-3} for $p = 1.2$ and 8.9×10^{-5} for $p = 1.1$.

p	h	nodes	$L^2(u)$	$L^2(\nabla u)$	residual
1.2	0.2	88	1.05×10^{-1}	3.04×10^{-1}	2.8×10^{-7}
	0.1	362	2.50×10^{-2}	1.40×10^{-1}	4.1×10^{-7}
	0.05	1452	6.26×10^{-3}	6.90×10^{-2}	2.3×10^{-7}
	0.025	5809	1.60×10^{-3}	3.37×10^{-2}	1.15×10^{-7}
1.1	0.2	88	2.74×10^{-1}	5.42×10^{-1}	3.9×10^{-6}
	0.1	362	7.63×10^{-2}	2.73×10^{-1}	2.5×10^{-6}
	0.05	1452	2.18×10^{-2}	1.37×10^{-1}	1.3×10^{-6}
	0.025	5809	5.96×10^{-3}	6.63×10^{-2}	6.3×10^{-7}

$p = 1.5$. We have no explanation, proved or heuristic, for the slow convergence of the iteration as $p \to 1$, beyond the fact that ∂F becomes multivalued in the limit.

6.4. **An x-dependent Young function: manufactured solution.** On $(0,1)^2$ with uniform meshes of $n \times n$ squares (two triangles each, $h = 1/n$) we take $G(x,t) = t^2/2 + (1 + x_1)t^4/4$, $u = e^{x_1+x_2}$, $f = -\operatorname{div}(g(x,|\nabla u|)\nabla u/|\nabla u|)$ computed symbolically, Dirichlet data u, a three-point load rule, $r = 100$ and tolerance 10^{-10}. Here (S) holds by Lemma 3.2 and Remark 4.2, and Theorem B(a) applies ($k(x,t) = g(x,t)/t = 1 + (1 + x_1)t^2 \geq 1$); Theorem C applies by Remark 4.1. Table 6 shows orders 2.00 and 1.00.

6.5. **The role of the parameter r.** By Theorem C the discrete iteration converges for every $r > 0$, but the speed depends strongly on r (Table 7). Heuristically, the best r is of the order of $g(t)/t$ at the typical size t of $|\nabla u|$: for the manufactured problem $|\nabla u|$ reaches $\sqrt{2}e^2 \approx 10.5$ and $g(x,t)/t = 1 + (1 + x_1)t^2$ is large, and $r = 100$ is best; for the dead core problem $r = 30$ is best. We have no theory for this. Table 7 also shows a caveat on the stopping criterion: for the dead core problem with $r = 100$ the error at termination (1.1292×10^{-3}) differs in the third digit from the value 1.1204×10^{-3} obtained with the other values of r, i.e. for large r a small primal residual

TABLE 6. Manufactured solution with x-dependent G, $r = 100$: relative true errors and pairwise orders.

n	nodes	iterations	$L^2(u)$	order	$L^2(\nabla u)$	order
10	121	163	7.16×10^{-4}	–	2.89×10^{-2}	–
20	441	190	1.77×10^{-4}	2.015	1.44×10^{-2}	1.001
40	1681	208	4.41×10^{-5}	2.004	7.22×10^{-3}	1.000
80	6561	218	1.10×10^{-5}	2.001	3.61×10^{-3}	1.000

TABLE 7. Iterations versus r. Left: manufactured problem, $n = 40$, tolerance 10^{-10} (relative L^2 error between 4.413×10^{-5} and 4.414×10^{-5} in all converged runs). Right: $e^t - 1$, $c = 8$, $h = 0.05$ (1452 nodes), tolerance 10^{-8}; relative L^2 error at termination and maximal discrete gradient on the triangles with barycentre in $\{|x| < 0.2\}$ (exact value 0). *: not converged in the maximal number of iterations (5000, resp. 6000).

manufactured		dead core ($e^t - 1$)			
r	iterations	r	iterations	$L^2(u)$	$\max_{\lvert x\rvert<0.2}\lvert\nabla u_h\rvert$
1	5000*	1	6000*	1.1204×10^{-3}	1.6×10^{-7}
10	1167	3	2139	1.1204×10^{-3}	1.1×10^{-7}
30	396	10	650	1.1204×10^{-3}	1.1×10^{-7}
100	208	30	244	1.1205×10^{-3}	9.9×10^{-8}
300	516	100	529	1.1292×10^{-3}	6.9×10^{-9}

does not guarantee a small iteration error (the term $\|\nu_n - \nu_{n-1}\|$ in (2.9), the dual residual, is not monitored). Monitoring both the primal and the dual residual, which is standard practice for ADMM [14], is advisable; we recommend it but have not rerun the experiments with it.

6.6. **Numerical check of the Lyapunov identity.** To test the mechanism of the proof of Theorem A we ran the discrete iteration on the disk mesh $h = 0.1$ (362 nodes) for L iterations and built a saddle-point proxy from the last iterate, $u^* = u_L$, $\nu^* = \nabla u_L$, $\eta^* = \eta_{L+1} + r(\nu_L - \nu_{L-1})$; by (2.7) these satisfy $\nu^* = \nabla u^*$ and $(\eta^*, \nabla w)_h = \ell_h(w)$ exactly up to round-off (relative residual $\le 4\times10^{-13}$), which is all that (2.9) uses, while $\eta^* \in \partial F_h(\nu^*)$ holds up to the final iteration error. We evaluated both sides of (2.9), the increments $E_n - E_{n-1}$, the cross term $r(s_n, \nu_n - \nu_{n-1})$ and m_n. Since (2.9) is an algebraic identity once $\nu^* = \nabla u^*$ and the multiplier equation hold, its verification to machine precision is a check of the implementation rather than of the theorem; the substantive content is the sign of the two terms that the proof requires to be nonnegative, which depend on $\eta_{n+1} \in \partial F_h(\nu_n)$. Table 8 reports these quantities over the iterations $n \le n_a$ in which $E_{n-1}/E_0 > 10^{-20}$ (beyond that, everything is at round-off level). In that range E_n is strictly decreasing and both terms are strictly positive in all five cases, including the nonsmooth case $e^t - 1$ with a multivalued ∂F. Over the full range $n \le L - 500$, E_n never increases by more than $3 \times 10^{-28}E_0$ and the two terms are $\ge -9\times10^{-28}E_0$, i.e. nonnegative up to round-off. Figure 3 shows the decay of E_n; it looks linear in all five cases, with rates depending strongly on G and r, which Theorem C does not predict (near $n \approx L$ the observed rate is affected by the use of a proxy taken from iterate L; the figure is cut at 10^{-24}). The original check `lyapunov_check.py` supplied with the code (which uses the last iterate itself as proxy) gives the same conclusions.

6.7. **A two-phase problem on a sector.** Finally we solve, on the unit disk with $f = 4$ and $u = 0$ on the boundary, the two-phase problem $G(x, t) = t^2/2 + a(x)t^6/6$ with $a = 0$ on the quarter disk $\{0 < \arg x < \pi/2\}$ and $a = 1$ elsewhere ($h = 0.03$, 4027 nodes, $r = 3$, tolerance 10^{-9}). The iteration converged in 60 iterations, in agreement with Theorem C; $\max u_h = 0.70897$, between the values

TABLE 8. Lyapunov identity (2.9) for the discrete iteration, disk mesh $h = 0.1$. All quantities are divided by E_0. The identity residual is the maximum over $n \le L-500$; the last three columns are over $n \le n_{\mathrm{a}}$, the iterations with $E_{n-1}/E_0 > 10^{-20}$ (negative increments mean that E_n strictly decreases).

G	r	L	n_{a}	max|identity res.|	$\max(E_n - E_{n-1})$	$\min r(s_n, \nu_n - \nu_{n-1})$	$\min m_n$
$t^2/2 + t^4/4$	1	1500	36	4.0×10^{-17}	-9.1×10^{-21}	2.1×10^{-21}	4.7×10^{-21}
$t^{1.5}/1.5$	1	2500	462	6.1×10^{-16}	-7.0×10^{-22}	8.6×10^{-25}	6.9×10^{-22}
$t^4/4$	1	1500	102	4.8×10^{-16}	-4.5×10^{-21}	1.7×10^{-22}	3.9×10^{-21}
$e^t - 1 - t$	1	1500	35	5.2×10^{-18}	-8.2×10^{-21}	2.0×10^{-21}	4.2×10^{-21}
$e^t - 1$, $c = 8$	30	2500	453	7.5×10^{-16}	-9.4×10^{-22}	2.0×10^{-24}	9.1×10^{-22}

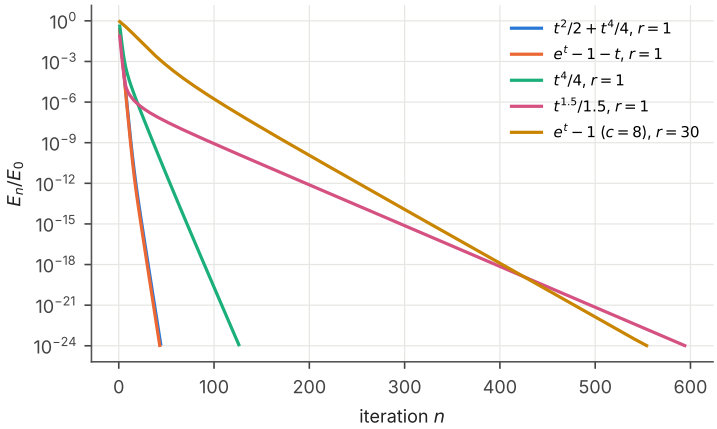


FIGURE 3. Decay of the Lyapunov functional E_n/E_0 of Theorem A for the discrete iteration on the disk mesh $h = 0.1$, plotted until E_n/E_0 reaches 10^{-24}.

for $a \equiv 1$ (0.66650, 52 iterations) and $a \equiv 0$ (0.99980, 64 iterations; exact value 1). Figure 4 shows the solution, the modulus of its gradient, which is larger in the sector where only the quadratic phase is present, and the coefficient as sampled at the barycentres. As explained in Remark 5.2, this experiment is qualitative: no exact solution is available, and neither (S) nor the absence of a Lavrentiev gap is guaranteed by the results we know, so we do not claim that u_h approximates the continuous minimiser.

6.8. **Comparison with the literature on rates.** For the p-Laplacian and related problems the proved a priori estimates for P1 elements are of order h in $W^{1,p}$-type norms or quasi-norms, under regularity assumptions on the solution: Barrett and Liu [10] (order h in $W^{1,p}$ for $p < 2$ and in $W^{1,q}$, $q \in [1, 2]$, for $p > 2$, for sufficiently regular solutions), Liu and Barrett [46] and Ebmeyer and Liu [30] (quasi-norm estimates), Diening and Růžička [29] and Diening and Kreuzer [28] (order h for $\|\mathcal{V}(\nabla u) - \mathcal{V}(\nabla u_h)\|_{L^2}$, $\mathcal{V}(a) = (\varphi'(|a|)/|a|)^{1/2}a$, when $\mathcal{V}(\nabla u) \in W^{1,2}$, for general N-functions of class $\Delta_2 \cap \nabla_2$); for $p(x)$, Del Pezzo and Martínez [27] (order $h^{p_1/2}$, and h under additional regularity, in $W^{1,p(\cdot)}$, $p \le 2$) and Breit, Diening and Schwarzacher [15] (order h^{α} in the quasi-norm for $p \in C^{0,\alpha}$). Glowinski and Marrocco [40] prove convergence without rate and $W^{1,p}$ rates for $u \in W^{2,p}$. The L^2 estimate in [46, Thm. 4.3] is obtained from the energy estimate by Hölder's inequality. We have found no proof of an $O(h^2)$ estimate in L^2 for the p-Laplacian or for the g-Laplacian: a duality (Aubin–Nitsche) argument is not available in general in the degenerate or singular case. The orders 2 in L^2 and 1 for the L^2 norm of the gradient observed in Tables 3–6 therefore go beyond the proved theory and are reported as observations only; they are consistent

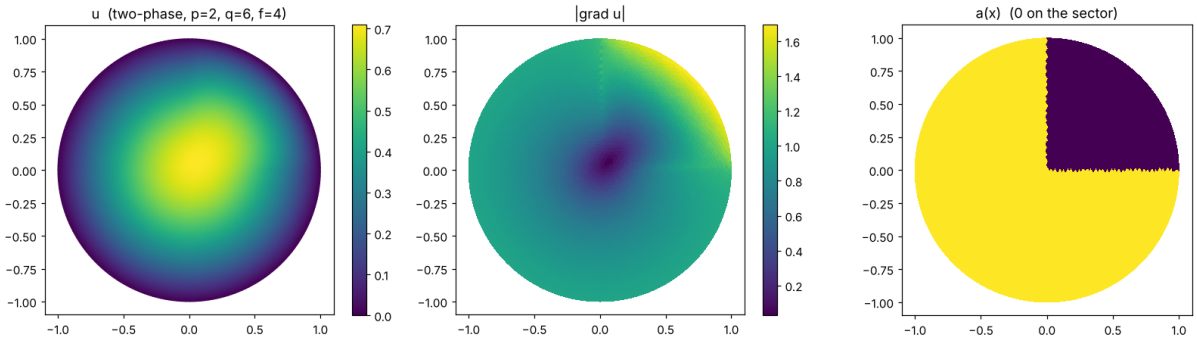


FIGURE 4. Two-phase problem $t^2/2 + a(x)t^6/6$ on the disk, $f = 4$, $a = 0$ on a quarter-disk sector. Left: u_h; centre: $|\nabla u_h|$ (piecewise constant); right: a sampled at barycentres.

with linear finite element heuristics when $g(x,t)/t$ is bounded above and below on the range of $|\nabla u|$, which is the case in our radial examples away from the origin (for several of our G, e.g. t^p/p with $p \neq 2$, $g(t)/t$ degenerates or blows up as $t \to 0$) and except in the dead core of $e^t - 1$. The reduced orders for the discontinuous coefficient are not covered by any estimate we know. The previous computations with the decomposition–coordination method that we are aware of report errors on a single mesh [4] or, for discontinuous Galerkin discretisations, L^2 errors on a sequence of meshes without fitting an order [26].

6.9. **Limitations.** (i) Orders are fitted over three meshes; the meshes are unstructured, so the pairwise orders fluctuate (Section 6.2), and the target h is used instead of the actual mesh size (Section 6.1). (ii) The disk is approximated by inscribed polygons; errors are computed on Ω_h. (iii) The iteration is stopped by the primal residual only; this does not control the iteration error uniformly in r (Section 6.5). (iv) Twelve runs did not reach the tolerance: the two finest meshes for $e^t - 1$ in Table 4, all eight runs for $p \in \{1.1, 1.2\}$ (Table 5), and the runs with $r = 1$ in Table 7; the corresponding errors may contain an iteration component (a few percent for $p = 1.1$, Section 6.3), and the fits for $e^t - 1$ and $p \in \{1.1, 1.2\}$ use these runs. (v) The "exact" radial solutions are computed numerically (Section 6.1). (vi) All test problems except the sector example have smooth or explicit solutions, so the experiments say nothing about the Lavrentiev regime. (vii) Iteration counts depend on h and r, and we have no theory for this.

7. Conclusions and open problems

For the decomposition–coordination method with $\rho = r$ applied to the g-Laplacian with an arbitrary convex Young integrand, we proved weak convergence at the continuous level under the assumption (S) that the augmented Lagrangian has a saddle point (Theorem A), strong convergence under uniform monotonicity properties that include $e^t - 1 - t$ and the p-Laplacian with $1 < p < 2$ (Theorem B), and convergence of the finite element iteration for every $r > 0$ and every superlinear convex G without any regularity assumption (Theorem C). For $p < 2$ the hypotheses of the classical theorem fail, while the method converges under (S); for $p > 2$ the applicability of the classical framework depends on how its differentiability hypothesis is read (Section 2.6). For the autonomous radial test problems we also proved convergence in energy of the P1 minimisers (Proposition 5.3); in general, convergence of u_h to u is not addressed. The hypothesis (S) is a regularity property of the solution; we reviewed what is known, and in particular that it is not available in general for exponential energies and for double phase problems up to the boundary. Open problems:

(1) Convergence of the continuous iteration when (S) fails, for instance for $1 < p < 2$ when the minimiser is not in H^1_0, or for $p > 2$ when $\nabla u \notin L^{2p-2}$; a formulation in $W^{1,p}_0 \times L^p \times L^{p'}$ would require a non-Hilbertian analysis.
(2) Strong convergence for general convex G (we do not know a counterexample within problem (1.1)), the case $\rho \neq r$, and rates for the iteration, including their dependence on h, p and r and a theory for the choice of r.
(3) Convergence of the P1 minimisers u_h to u beyond the autonomous radial case of Proposition 5.3 (for instance for discontinuous coefficients, as in the two-phase tests), and a priori error estimates explaining the observed $O(h^2)$ behaviour in L^2 for degenerate or singular G, and for G without Δ_2.
(4) Lavrentiev-aware discretisations, for instance a Crouzeix–Raviart version of the method in the spirit of [7], for which an analogue of Theorem C would hold with the same proof.
(5) Global regularity, hence (S), for Musielak–Orlicz $G(x,t)$ without Δ_2, and for double phase energies up to the boundary.

Software availability. The Python package `glaplace` (NumPy/SciPy; MIT license), which implements the method for arbitrary Young functions on general two-dimensional triangulations and contains the scripts for all experiments of Section 6, is available at `https://github.com/jfbonder/glaplace`. It generalises the code for the $p(x)$-Laplacian of [4].

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